

Graph Signal Processing, Graph Neural Network and Graph Learning on Biological Data: A Systematic Review

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Abstract—Graph networks can model data observed across different levels of biological systems that span from population graphs (with patients as network nodes) to molecular graphs that involve omics data. Graph-based approaches have shed light on decoding biological processes modulated by complex interactions. This paper systematically reviews graph-based analysis methods of Graph Signal Processing (GSP), Graph Neural Networks (GNNs) and graph topology inference, and their applications to biological data. This work focuses on the algorithms of graph-based approaches and the constructions of graph-based frameworks that are adapted to a broad range of biological data. We cover the Graph Fourier Transform and the graph filter developed in GSP, which provides tools to investigate biological signals in the graph domain that can potentially benefit from the underlying graph structures. We also review the node, graph, and interaction oriented applications of GNNs with inductive and transductive learning manners for various biological targets. As a key component of graph analysis, we provide a review of graph topology inference methods that incorporate assumptions for specific biological objectives. Finally, we discuss the biological application of graph analysis methods within this exhaustive literature collection, potentially providing insights for future research in biological sciences.

Index Terms—Graph signal processing, graph neural network, graph convolutional network, graph learning, biological data.

I. INTRODUCTION

THE proliferation of multimodal technologies emerging in past decades have accumulated diverse and large volumes of biological data that are increasing exponentially each year. One major category of biological data is the “omics”, which implies a comprehensive or complete profile of a set of molecules [1]. The first omics discipline, genomics, was to sequence the human genome, which was a titanic task that generated a massive amount of data. It has been available for two decades and has enormously enhanced our understandings of complex diseases and traits [2]. Following on from genomics, various other types of omics data spanning all levels of biology, including RNA molecules, proteins, metabolites, cellular lipids, epigenome, and microbiome, have been generated efficiently, with low-cost thanks to improved biotechnology [3]. Another category of biological data is physiological monitoring data measured by wearable sensors or monitors in hospital and home settings, which can continuously track various biosignals from the human body or other organic tissues, such as the heart, brain, muscle, blood and adipose tissue [4]. One of the first sets of physiological data applied in clinical diagnosis was electrocardiography (ECG), mapping the cardiac electrophysiological function over time as heartbeats by detecting electrical changes in cardiac structures induced by cardiac muscle depolarization [5]. Many different types of biological activities can now be tracked, such as electrodermal activity, cardiovascular data, respiration, muscular and skeletal movement and positioning, which can be gauged by a wide variety of techniques including ECG, electromyography (EMG), intracranial/scalp electroencephalography (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), functional near-infrared spectroscopy (fNIRS) and so on [4], [6]. The vast amount biological omics and physiological data now available have allowed us to expand our understanding of physiological and pathological processes and develop novel healthcare solutions for complex diseases [7]. However, it also imposes major analysis challenges

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due to its complexity and size. Real-world biological systems are seldomly self-regulated. Instead, the biological data tend to be generated by complex modulations across multiple systems. Thus, classical signal processing methods in the time and frequency domain are insufficient to decode regulations across various biological targets. Complex networks are an emerging tool to describe essential features of interactions of biological targets [8], [9]. Besides, it provides abundant analysis methods from a network viewpoint that can characterize complexities of biological data [10]. A graph network is a collection of vertices that are connected by a set of edges. Graph networks are a typical data structure in biology, which model interactions across various biological objects. It has attracted significant interests in the study of physiological and disease states with perspectives of network interaction. For instance, the complexity of the human brain relates to intricate anatomical and functional connectivity networks, analyses of which have attracted massive attention in the field of cognition, psychiatry, psychology, neurology, and neuroscience [11]. Moreover, it is critical for advances of healthcare to investigate molecular interactions, regulatory networks, and functions of molecules at different organizational levels to better understand health and disease states [12]. Emerging graph signal processing (GSP) approaches, such as Graph Fourier Transform (GFT) and graph filters, have been exploited for analyzing graph-structured data [13]–[15] that would potentially benefit investigations of complex biological networks. Deep learning with convolutional neural networks (CNNs) has achieved success for analyses of regularly structured data, such as images and speech [16]. With the development of GSP, graph neural networks (GNNs) were proposed to adapt to graph-structured data, which can be particularly useful for analyzing big datasets of biological networks that have been accumulated and currently keep growing exponentially. The graph topology is fundamental for data analysis in the graph domain. The underlying graph network is available in some cases, such as brain structural connectivity networks generated by diffusion magnetic resonance imaging (MRI) [17]–[19]; interactions across omics data generated from wet lab experiments or curation of biomedical literature [1], [3]. However, network interactions across biological targets are often not available and have to be inferred from observed data. The constructed graph topology directly influences results of GSP and graph-based neural networks. Thus, it is critically important to learn graph network topologies that are adaptive to specific objectives.

Graphic approaches developed recently have great potential to improve network structured data analysis in non-Euclidean space. Methodological reviews have been published for the development of GSP methods [13]–[20], [21], GNNs [22], [23], and graph topology learning approaches [24]. Reviews of biological applications of these aforementioned graph-based analysis methods are still limited and tend to focus on one particular biological objective, summarized in Table I. The methodologies and their specific applications for surveys are listed in the second and the third column of Table I.

Our survey presents an exhaustive literature review about the application of GSP, GNNs, and graph inference learning approaches on comprehensive biological data. We present study

TABLE I
SURVEYS OF THE APPLICATIONS OF GNN AND GRAPH INFERENCE
LEARNING ON BIOLOGICAL DATA

Ref.	Methodology	Application
[25]	GNN	Precision oncology
[26]	Deep learning	Drug discovery
[27]	Graph- and rule-based learning	Cancer-gene interaction
[28]	GNN	Drug discovery
[29]	GSP	Functional brain imaging
[30]	Graph inference learning	Functional brain connectivity
[31]	Deep learning	Biological networks

selection criteria for this systematic literature review in Appendix 1 Study Selection Criteria. With the collected literature, we review the current progress of this area of knowledge and discuss the possible research directions for biological network analyses. In summary, this review makes contributions in the following five aspects:

- Algorithms and formulations of GSP, GNNs, and graph topology inference methods for biological data. We present the mathematic equations and models that have been applied to biological data.
- Review of the application of GSP approaches in biology. This is the first article that performs an exhaustive literature review about applying GSP methods in biological networks.
- Categorize applications of GNNs at different levels for various biological objectives. We classify biological applications of GNNs based on different network learning targets: node, edge, and graph. We review biological objectives, architecture designs of GNNs, and particular technical challenges for each type of implementation.
- Summarize graph learning methods with biological objectives. We formulate the primary graph topology inference methods and summarized graph learning studies with specific biological assumptions. Particularly, we further discuss studies that infer graph topologies using GNNs.
- Future directions. We discuss some open issues to be addressed for the application of graphic data analysis methods in biological research.

The remainder of this paper is organized as follows: the relational across different graphic methods we cover in this review is given in Section II. Reviews of studies that analyzed biological data in graph domain using graph signal processing approaches and GNNs are shown in Section III and Section IV, respectively. Section V provides a review of graph topology inference methods applied to biological data. Section VI presents a discussion of the collected studies in this survey and gives future research directions.

II. OUTLINE AND CATEGORIZATION

We performed a comprehensive literature review about the application of GSP, GNN, and graph learning methods in biological data. We illustrate the outline of methodologies and categorizations of their biological applications in Fig. 1.

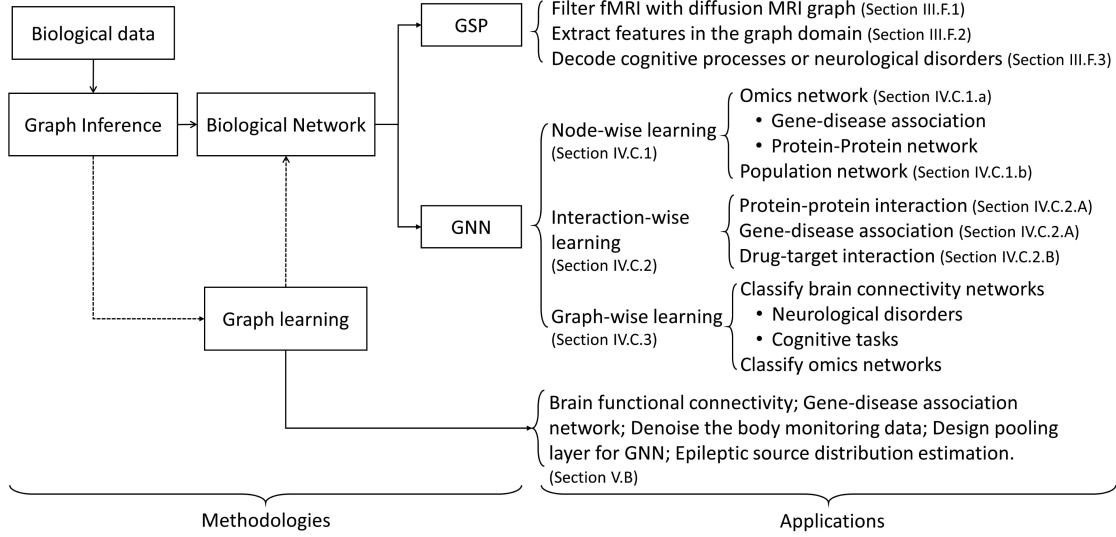


Fig. 1. Methodology outline and application categorization.

GSP and GNN are two classes of graph-based analysis approaches that can be applied to analyze biological networks. A primary application of GSP methods is to filter biological signals with underlying or predefined graph structures, which generally fall into three categories: filtering fMRI signals with diffusion MRI graphs, feature extractions in the graph domain, and decoding cognitive processes or neurological disorders. GNNs are the graphic machine learning paradigms and can be performed for different biological targets with network graph, node, and interaction oriented learning procedures.

Graph inference methods can be exploited to generate interaction networks across biological targets when the underlying biological networks are not available. Sophisticated inference methods have been proposed to infer networks with assumptions for specific graph structures or particular biological objectives.

III. GRAPH SIGNAL PROCESSING

Graph signal processing (GSP) is a collection of classical signal processing techniques that are adapted to graph-structured data. In this section, we first provide important GSP equations and algorithms that have been applied to analyze biological data. The GSP operations included are the graph Fourier transform (GFT), filters in graph domain, the graph convolution and shift operation, and the vertex-frequency representation. In the last section, the application of these GSP techniques in biological studies is reviewed, mainly in three categories: decode fMRI with structural connectivity networks using various graph filters, feature extractions in the graph domain for biomedical engineering, and interpretations of cognitive processes or neurological disorders.

A. Preliminary

We can denote an undirected graph as $G = (V, E, \mathbf{A})$, where V is a set of vertices with $|V| = M$, E is a set of edges (i, j) connecting vertices $i, j \in V$, M is the number of vertices, and $\mathbf{A}$ is the adjacency matrix whose nonzero elements a_{ij} are weights

of edges in $(i, j) \in E$. The Laplacian matrix of a graph G is $\mathbf{L} = \mathbf{D} - \mathbf{A}$. $\mathbf{D}$ is a degree matrix, which is a diagonal matrix with $d_{ii} = \sum_{j=1}^M a_{ij}$. We normalize the Laplacian matrix by the degree matrix, i.e., $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}}$, which is referred to as the normalized Laplacian matrix. The adjacency matrix, the Laplacian matrix and their normalized counterparts are known as graph shift operators that satisfy $\mathbf{A}(i, j) = 0$ and $\mathbf{L}(i, j) = 0$ when $(i, j) \in E(i, j)$ [13].

B. Graph Fourier Transform (GFT)

The GFT is analogous to the Fourier Transform of time series that can decompose graphs onto a variety of graph frequency basis, which indicate various levels of smoothness [14]. The graph spectral representations can be generated with the eigenvalue decomposition of the adjacency matrix, the Laplacian matrix and the normalized Laplacian matrix. They have different properties and illustrate discrete perspectives of a graph, which have been investigated in an article [32]. The Laplacian matrix and the normalized Laplacian matrix can be applied to undirected graphs. They are symmetric and positive semidefinite that can prevent a number of analytical and computational problems caused by using the adjacency matrix [13]. We give the graph spectral analysis based on the eigenvalue decomposition of the normalized Laplacian matrix in (1), which is commonly used in the literature of biological graph networks.

$$\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^{-1} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T, \quad (1)$$

where $\mathbf{\Lambda}$ is the diagonal matrix of the eigenvalues with an ascending order $\text{diag}(\lambda_1, \dots, \lambda_M)$, and the columns of $\mathbf{U}$ are the corresponding eigenvectors $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_M]$. The eigenvectors of a graph are the basis of the graph Fourier transform, which can also be called eigenmode.

Generally, any form of transforming a graph and its corresponding vertices that employs the eigenvectors of the graph Laplacian can be considered as a Laplacian eigenmaps. One example of the eigenvalue decomposition of the graph network

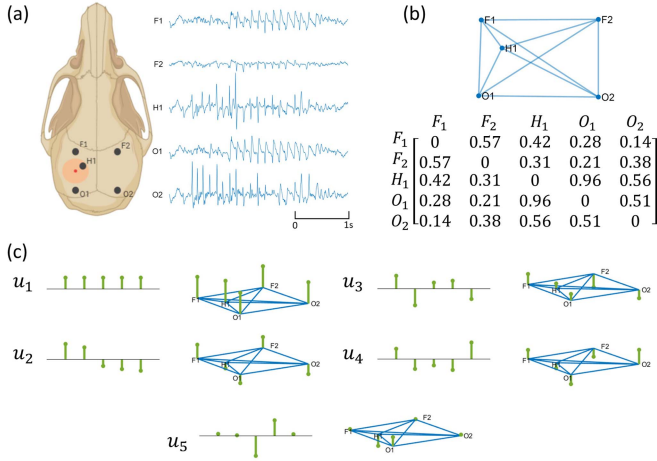


Fig. 2. Laplacian matrix eigenvalue decomposition of a graph network constructed from 3.36 s EEG recordings of a rodent brain. (a) The EEG electrodes placement and recordings of the rodent brain. (b) The graph network across five electrodes: the correlation coefficients. The five eigenvectors u_1, u_2, u_3, u_4 , and u_5 shown in (c) were generated from the Laplacian matrix of the network in (b), and they were ranked by the eigenvalues with the ascending order. For each eigenvector, its value distributions are demonstrated with the sensor index and the sensor locations, respectively.

is illustrated in Fig. 2. The EEG data were recorded from a traumatic brain injury rodent model [33]. The placement of the implanted electrodes and one recorded spike-wave discharges are shown in Fig. 2(a). We generate the graph network across the five electrodes using the correlation coefficients illustrated in Fig. 2(b). The graph eigenvalue decomposition was performed on the Laplacian matrix of the EEG correlation matrix. The eigenvectors in Fig. 2(c) were ranked by their corresponding eigenvalues in the ascending order. The smallest eigenvalue corresponds to the eigenvector with the constant elements. Additionally, we can see the association between the magnitude of eigenvalues and the smoothness of the eigenvector distributed across the graph network.

C. Graph Filter

The filter in the graph domain can be developed based on the GFT basis. For data $\mathbf{x}$ that reside on vertices of a graph with dimension $M \times N$, where N is the sample size of the data. The GFT of the data $\mathbf{x}$ is $\mathbf{X} = \mathbf{U}^T \mathbf{x}$ and the inverse GFT is $\mathbf{x} = \mathbf{U} \mathbf{X}$. Consider a graph-domain filter $H(\Lambda)$, the output spectral of the graph signal $\mathbf{x}$ can be expressed as

$$\mathbf{Y} = H(\Lambda) \mathbf{X}. \quad (2)$$

The multiplication in graph spectral domain shown in (2) corresponds to convolution in vertex domain, which can be derived by taking the inverse GFT on (2) as [20]

$$\mathbf{y} = \mathbf{U} \mathbf{Y} = \mathbf{U} H(\Lambda) \mathbf{X} = \mathbf{U} H(\Lambda) \mathbf{U}^T \mathbf{x}. \quad (3)$$

Specifically, the filtered graph signal $y(n)$ on vertex n can be computed with this convolution operation as shown in

$$y(n) = \sum_{k=0}^{M-1} X(k) H(k) u_k(n), \quad (4)$$

where $u_k(n)$ is the n th element of the eigenvector $\mathbf{u}_k$ of the Laplacian matrix.

The eigenvalue decomposition of $\mathbf{L}$ requires M^3 arithmetic operations, which is not applicable for large-scale graphs. The spectral kernel $H(\Lambda)$ can be approximated with the Chebyshev polynomial $P(\Lambda)$ [20]. Thus, (3) can be expressed with the polynomial approximation as

$$\begin{aligned} \mathbf{y} &= \mathbf{U} H(\Lambda) \mathbf{U}^T \mathbf{x} \approx \mathbf{U} P(\Lambda) \mathbf{U}^T \mathbf{x} \\ &= \mathbf{P}(\mathbf{L}) \mathbf{x} = \sum_{k=0}^K h_k \mathbf{L}^k \mathbf{x}, \end{aligned} \quad (5)$$

where $\mathbf{P}(\mathbf{L}) = \sum_{k=0}^K h_k \mathbf{L}^k$ is the polynomial form of the kernel function $H(\mathbf{L})$, and K is the approximation order of the polynomial. Also, (5) is the graph convolution defined with a shift-and-sum operation.

D. Vertex-Frequency Representations

The GFT can analyze behaviours of a whole network in graph frequency space [15]. However, local characteristics of graph signals can be of interest in practice. The vertex-frequency representation corresponds to the time-frequency representation for time series analysis. We can denote the vertex-frequency representation of a graph signal that is localized to the vertex m around the k th spectral index as $S(m, k)$. It can be obtained by projecting a graph signal $x(n)$ onto a vertex-frequency kernel function $\mathcal{H}_{m,k}(n)$ as

$$S(m, k) = \sum_{n=0}^{M-1} \mathcal{H}_{m,k}(n) x(n). \quad (6)$$

There are two ways to generate the vertex-frequency kernel function. One way is to shift the graph signal $x(n)$ towards the vertex m , and then transfer it to the k th spectral index, as given by (7).

$$\begin{aligned} \mathcal{H}_{m,k}(n) &= \left(\sum_{p=0}^{M-1} H(p) \Delta(p) u_p(n) \right) u_k(n) \\ &= \left(\sum_{p=0}^{M-1} H(p) u_p(m) u_p(n) \right) u_k(n), \end{aligned} \quad (7)$$

where $\Delta(p) = \sum_{n=0}^{M-1} \delta_m(n) u_p(n) = u_p(m)$ is the GFT of a delta function $\delta_m(n)$ located at vertex m , expressed as

$$\delta_m(n) = \begin{cases} 1, & \text{for } m = n \\ 0, & \text{for } m \neq n. \end{cases} \quad (8)$$

The term in brackets of (7) is the graph shift performed with the graph delta function using the graph convolution operation, and $u_k(n)$ is used to localize to the k th spectral index.

Another way to generate the vertex-frequency kernel function is to firstly create a bandpass transfer function $H_k(\lambda_p)$ centered at the k th spectral index in the spectral domain, and then shift the graph towards vertex m using the graph convolution operator with the GFT of the delta function $\Delta(p)$ as given by

$$\begin{aligned}\mathcal{H}_{m,k}(n) &= \sum_{p=0}^{M-1} H_k(\lambda_p) \Delta(p) u_p(n) \\ &= \sum_{p=0}^{M-1} H_k(\lambda_p) u_p(m) u_p(n).\end{aligned}\quad (9)$$

1) Graph Wavelet Transform: The Graph wavelet transform was developed to balance the resolution trade-off in the vertex and frequency domain. We can define a graph wavelet in the spectral domain $H(\lambda)$ with the scaling parameters $s_i, i = 1, 2, \dots, K$. The graph wavelet kernel function can be used to replace the term $H_k(\lambda_p)$ in (9) with the scaled wavelet $H(s_i\lambda)$, as given by

$$\mathcal{H}_{m,s_i}(n) = \sum_{p=0}^{M-1} H(s_i\lambda) u_p(m) u_p(n). \quad (10)$$

The key of the graph wavelet transform is to design appropriate wavelet kernels and scaling frameworks. It is essential to develop tight wavelet frames, for which the sum of squares of all kernels is constant throughout the spectrum. The tight frame is necessary for stable signal reconstruction and efficient computations [34]. Different graph wavelets have been proposed to adapt to graph signals [15].

We show two signal adapted graph wavelet frames, Spline and Warped Translate wavelet kernels, in Fig. 3. The graph network and signals used in this figure were diffusion MRI and resting-state functional MRI of one participant in the human connectome project [35]. The Spline wavelet frame provides a smooth transition across the graph spectrum range, which is defined by the eigenvalues of the graph Laplacian matrix [36]. As shown in Fig. 3(a), the Spline kernels can cover the whole spectrum range limited by the maximum eigenvalue 40.19 of the graph network.

Warped Translates can not only adapt to the maximum eigenvalue to cover the spectrum range but also can adapt to the distribution of eigenvalues [37]. Specifically, it can concentrate wavelet kernels in the spectrum range where the eigenvalues are densely located. As shown in Fig. 3(b), the distribution of Spline kernels with Warped Translates is aligned with the density distribution of eigenvalues shown in Fig. 3(c).

2) Graph Slepian: The graph Slepian is an alternative approach to the graph wavelet transform that can concentrate energy in the band-limited frequency spectrum and the predefined subgraphs simultaneously [38]. The eigenvectors with the smallest N_w eigenvalues are selected for the band limitation, and the band-limited graph spectrum is $\mathbf{U}_W = \mathbf{U}\mathbf{W}$, where $\mathbf{U}$ has eigenvectors of Laplacian matrix as columns, and $\mathbf{W}$ is the linear combination matrix of the eigenvectors corresponding to

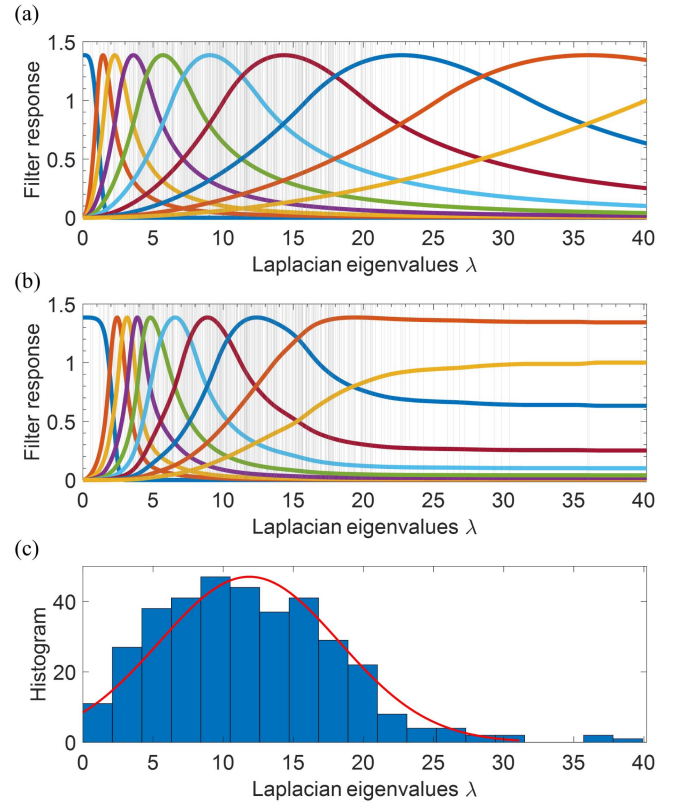


Fig. 3. Signal-adapted graph wavelet frameworks. (a) Spline-based wavelet kernels. (b) Warped Translate wavelet kernels. (c) Histogram distribution of Laplacian eigenvalues of a graph signal. Vertical lines in (a) and (b) indicate the locations of eigenvalues.

the smallest N_w eigenvalues. The graph Fourier basis representation of a signal without band-limitation is $\mathbf{y} = \mathbf{U}\mathbf{X}$, while the band-limited signal representation is $\mathbf{y}_U = \mathbf{U}\mathbf{W}\mathbf{X}$.

The diagonal matrix $\mathbf{S}$ was applied to select predefined subnetwork, where $S_{k,k} = 0/1$ indicates the exclusion/inclusion of a node for the subnetwork selection. Thus, the band-limited and subnetwork concentration matrix can be expressed as

$$\mathbf{C} = \mathbf{W}^T \mathbf{U}^T \mathbf{S} \mathbf{U} \mathbf{W}. \quad (11)$$

The Rayleigh quotient can obtain the optimal combination of the band-limited frequency spectrum in a predefined subnetwork, as given by

$$\mu = \frac{\mathbf{X}^T \mathbf{C} \mathbf{X}}{\mathbf{X}^T \mathbf{X}}. \quad (12)$$

The Slepian design has been further generalized to substitute selection binary matrix $\mathbf{S}$ with an importance weighted matrix [39], considering both energy concentration and the optimization of local frequency on the graph.

E. Graph Fourier Transform for Directed Graphs

The eigendecomposition of the Laplacian matrix can generate GFT basis for undirected graphs as illustrated in Section III-C. However, the GFT eigenbasis of a Laplacian matrix may not be valid for directed graphs. Since eigenvectors of directed

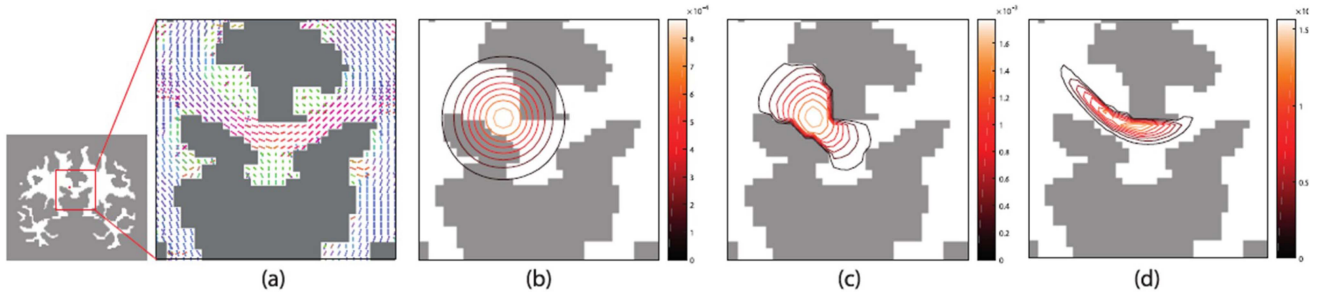


Fig. 4. Deconvolution of fMRI with brain structural connectivity graph. The contour profiles of spatial filters with different adaptations to the heterogeneous orientation distribution of white matter fibers are shown in (a). (b) The isotropic Gaussian filter without adaptations to anatomical morphology nor the underlying white matter structure. (c) A filter adapted to the anatomical morphology, but not the white matter structural orientations. (d) A graph filter adapted to both the anatomical morphology and the underlying structure (cite Fig. 1 in [41], copyright has been approved by IEEE).

graphs can show high multiplicity that a set of orthogonal GFT basis is not unique, and lead to divergent results. In addition, the adjacency matrix of a directed matrix displays eigenvalues with multiplicity higher than one that the adjacency matrix is not guaranteed to be diagonalizable.

A more general GFT method that is applicable to directed graphs is the Jordan decomposition of the adjacency matrix, which defines the associated generalized eigenvectors as the GFT basis [14].

$$\mathbf{A} = \mathbf{V}\mathbf{J}\mathbf{V}^{-1}, \quad (13)$$

where $\mathbf{J}$ is the Jordan normal form, and $\mathbf{V}$ is the matrix of generalized eigenvectors. The details of Jordan decomposition can be found in [14]. However, the Jordan generalized eigenbasis is limited because of its well-recognized instability of computation. An alternative approach has been proposed to compute a set of orthonormal vectors that minimize the total variation through the Lovász extension of the cut size [40]. Meanwhile, the orthogonality of GFT basis is also preserved. This optimisation GFT can be formulated as

$$\begin{aligned} \min_{\mathbf{V} \in \mathbb{R}^{M \times M}} \text{GDV}(\mathbf{V}) &:= \sum_{k=1}^N \text{GDV}(\mathbf{v}_k) \\ \text{s.t. } \mathbf{V}^T \mathbf{V} &= \mathbf{I}, \mathbf{v}_1 = c\mathbf{1}, \end{aligned} \quad (14)$$

where GDV represents the Graph Directed Variation, $\mathbf{V}$ is the set of orthonormal vectors, $\mathbf{v}_1$ is the first basis vector that is composed of constants $c = 1/\sqrt{M}$ with no variations. Besides its compatibility to directed graphs, it is particularly adaptable to graph signals with clusters that closely connected vertices show smooth variations, and vertices from different clusters exhibit arbitrary variation.

F. Applications

GSP provides us with tools to analyze structured but irregular data, graph networks. In neuroscience, brain activities are associated with complex functional and structural connectivity networks. The analysis of brain networks shed light on our understanding of neurological disorders, cognitive processes, and

TABLE II
SUMMARY OF BRAIN STRUCTURAL GRAPH FILTERS APPLIED TO fMRI DATA

Graph filters		Reference
Eigenvectors	$\mathbf{y} = \mathbf{U}^T \mathbf{x}$	[42]–[44] (Adjacency matrix) [19] (Laplacian matrix)
Polynomial approximation	$\mathbf{y} = \mathbf{P}(\mathbf{L})\mathbf{x}$	Heat kernel [41] Spline kernel [17] Diffusion process [45]
Energy concentration	$\mathbf{y} = \mathbf{W}^T \mathbf{x}$	Slepian basis [46], [47]

brain-computer interface design. The literature of applications of GSP on biological data are reviewed as below:

1) Decode Functional MRI With Brain Structural Graph:

Functional magnetic resonance imaging (fMRI) maps brain functional activities by measuring the blood-oxygen-level-dependent (BOLD) contrast. fMRI is a non-invasive imaging modality that holds rich information provided by the underlying anatomy.

Isotropic Gaussian filtering is commonly used to enhance signal-to-noise ratio (SNR) of fMRI signals. The assumption of isotropy is violated by the spatially correlated structure of the BOLD signal, which is related to the diffusion tensor¹ regarding morphology and direction. One example of the isotropic anatomical distribution of the brain structure is illustrated in Fig. 4 that the orientation distribution of white matter fibers is shown in panel (a). The spatial filters with various adaptations to the brain anatomical morphology and structure are illustrated in the (b) – (d) of Fig. 4.

We summarize graph filters applied to fMRI data with the adaptations to the brain structural distributions in Table II. A pioneering study [42] applied a graph filter generated from the brain structural network onto the functional signals. The graph filter was calculated directly from the adjacency matrix. They observed that the high-frequency eigenvectors are correlated with the greater cognitive flexibility of an individual. A similar graph filter diagram has been constructed from the brain structural network, which were further used to decode the brain graph

¹Diffusion tensor can image the cellular structures by measuring the diffusion of water molecules.

changes of athletes with concussions [43]. There were 22 regions of interest (ROIs) with BOLD signals filtered by low pass and high pass graph filters significantly different between the healthy controls and athletes with concussions. The positive correlation between the age and the high-frequency graph filtered energy in the lingual region of the right hemisphere was revealed in the athlete group, which was not shown in the healthy control group. Following this work, they performed a sliding-window based dynamic connectivity analysis across the ROIs selected by the graph frequency filters [44]. The durations of multiple functional connectivity states for the two groups of participants were significantly different.

It is also straightforward to use the eigenvectors of the Laplacian matrix of the brain structural graph to filter fMRI data. It was applied to project the fMRI statistics (mean, standard deviations, and variance) onto the structural MRI graph using the eigenvector of the Laplacian matrix, which can classify health control and Autism Spectrum Disorder (ASD) patients better than the statistics without being graph filtered [19].

The computation of the eigenvectors of the Laplacian matrix can be infeasible for the large-scale graphs. To deal with this issue, the polynomial approximation of the kernel function as given in (5) was applied to compute the graph filter efficiently. As summarized in Table II, the fMRI data was processed by the polynomial approximated graph filters in [17], [41] and [45] with the Spline kernel, the heat kernel and the diffusion process, respectively. In [41], the polynomial approximation of the heat kernel graph filter generated from the diffusion MRI, measuring the diffusion tensor, was applied to decode BOLD activations. It showed greater sensitivity and specificity in the activation mapping compared with the isotropic filtering.

In [17], patient-specific and voxel-wise cerebral hemisphere cortex (CHC) graphs were designed embedding abundant morphological properties. Since the voxel-wise CHC graphs have very high resolution with the order of 10^5 voxels, the energy decomposition of CHC graphs using eigendecomposition becomes computationally impractical. The spline-type polynomial was applied to approximate the graph spectral content. This work analyzed fMRI data with seven functional tasks, and showed that the BOLD signal energy decomposed with CHC graphs can capture task-specific spatial patterns of brain function activities.

It is worth noting the difference of graph filters between the region-wise and voxel-wise brain structural networks. Studies [19], [42]–[44] can generate graph filters from brain structural networks among ROIs using eigendecomposition method. However, this approach is computationally practical for voxel-wise brain structural networks, which are usually generated from high-resolution imaging modalities leading to networks with a large number of nodes. To deal with this issue, graph filters can be generated from the polynomial approximations, as shown in (5). Studies [17], [36] that considered voxel-wise networks applied graph filters with polynomial approximations.

Similar to the decode functional signals from structural connectivity network, a method was proposed to infer the brain structural connectivity network from the functional network with an assumption that the recorded fMRI was generated by a physical process on the structural graph [45]. Thus, the

functional network calculated from fMRI data and the brain structural network shared the same eigenvectors. They tested the deconvoluted structural connections between health controls and the attention deficit hyperactivity disorder (ADHD). The proposed approach can reveal the previously discovered highly connected brain regions of the ADHD group.

Kernel functions, including the Slepian basis and the Warped Translates, have been proposed to concentrate energy onto specific graph nodes or predefined subgraphs. The graph filter with localized energy concentration applied to fMRI studies are summarized in the last row of Table II. The Slepian basis was applied to construct graph filters adapted to diffusion MRI to focus energy on the predefined subnetworks. Results showed that energy localization could reveal local information used to be covered by the global organization [46]. Moreover, Bolton *et al.* [47] showed the Slepian kernel and localized GFT that enable spatial and graph spectral localization to classify fMRI data tasks better than the traditional GFT. Later on, they recovered brain activities from the BOLD signal with regularisation of both temporal and spatial variables [18].

2) Feature Extractions From Biological Graph Network:

Properties of grid-like regular data have been comprehensively studied using classic signal processing methods in both time and frequency domains. The characteristics of graph-structured data can be analyzed using GSP. In the literature of biological data, GSP has been applied to extract graph-based features from networks constructed from unimodal or multimodal biological data. Studies about the feature extraction in the graph domain are summarized in Table III, with the graph network constructed and the graph-based features given in the third and fourth columns. The graph-domain features mainly included signal or signal energy decomposed in the graph sub-frequency band [48]–[55], signal smoothness across underlying graph structure [56], and eigenvalues of graph Laplacian matrix [57].

Brain-computer interface (BCI) that wires the brain activities to external devices is an important research topic in biomedical engineering, potentially helping disabled people communicate with the environment. Still, further improvement of BCI techniques is required to benefit peoples' life reliably [62].

The most straightforward feature extraction or dimensionality reduction using GSP methods is to project a signal onto the lower/higher graph frequency band, corresponding to eigenvectors of the graph Laplacian matrix with the smaller/larger eigenvalues. In general, more signal power tends to concentrate on the graph lower frequency band. Two studies projected recorded scalp EEG [50] and MEG data [53] onto the lower and higher graph frequency band to reduce the data dimensionality for the following BCI and visual stimuli classification tasks, respectively. In addition to dimensionality reduction, the signal power in the graph frequency band was used as features for the BCI tasks classification. In [48], the signal power in each graph frequency band was calculated as features, from which more separable graph band powers were selected for the classification tasks.

Besides the eigenvectors of the Laplacian matrix, a novel subspace was proposed to produce the most separable projections for two BCI tasks. The projection of whitened EEG

TABLE III
SUMMARY OF FEATURE EXTRACTION IN THE GRAPH DOMAIN

Ref.	Recording modality; Application;	Graph network	Graph-based features extraction
[50], [53]	Scalp EEG; MEG; BCI.	EEG electrodes proximity and correlation coefficient	Dimensionality reduction: Signal projected onto the lower [50] / higher [53] graph frequency band. $\mathbf{f} = \mathbf{U}_l^T \mathbf{x}$ or $\mathbf{f} = \mathbf{U}_u^T \mathbf{x}$
[48]	Scalp EEG; BCI.	Cross frequency coupling across EEG electrodes	Graph-based features: The graph signal power projected onto graph basis of eigenvectors; Feature selection: The power on the graph-basis exceeding a threshold. $\mathbf{P}_{\text{GSP}}(\lambda_m)$ $= \frac{1}{T} \sum_{t=1}^T \mathbf{U}^T(\lambda_m) \mathbf{x} ^2$
[51]	Scalp EEG; BCI.	Spatial covariance matrix of electrodes	Graph-based features: Signal projected onto eigenvectors corresponding to the largest and smallest eigenvalues of the whitened covariance matrix. $\mathbf{f} = \begin{bmatrix} \mathbf{B}_{\lambda_0}^T \mathbf{P} \mathbf{x} \\ \mathbf{B}_{\lambda_{\max}}^T \mathbf{P} \mathbf{x} \end{bmatrix}$
[58]	fMRI; ASD classification.	Inverse of distances across ROIs	Graph-based features: GFT coefficients of fMRI data projected onto a matrix that simultaneously diagonalise the whitened data covariance in the graph space. $\mathbf{X}_t = \mathbf{U}^T \mathbf{x}_t$ $\mathbf{Z}_t = \mathbf{P}_{\text{SFM}} \cdot \hat{\mathbf{X}}_t$
[56]	Scalp EEG; Brain injury states.	Gaussian weighted electrodes distance	Graph-based features: Total variance (TV): the graph signal smoothness. TV $= \sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} w_{i,j} (x_i - x_j)^2 \right)^{1/2}$
[49]	iEEG; Seizure detection.	Data covariance with time lags	Graph-based features: • The energy in different graph frequency bands; • Min, max and average of eigenvalues; • The maximum and minimum frequencies of graph wavelet transform; • Graph Laplacian quadratic form. $\mathbf{P}_{\text{GSP}}(\lambda_m)$ $= \frac{1}{T} \sum_{t=1}^T \mathbf{U}^T(\lambda_m) \mathbf{x} ^2$
[54]	fMRI; Different age groups.	Functional data correlations across ROIs	Graph-based features: • Low-frequency graph components; Along with topological features of the correlation network: • Degree and betweenness centrality; • Clustering coefficient. $\mathbf{f} = \mathbf{U}_l^T \mathbf{x}$
[52]	EEG; Epilepsy detection.	Visibility graph of EEG time series	Graph-based features: The energy in different graph frequency bands. $\mathbf{P}_{\text{GSP}} = \sum_{m=0}^M \mathbf{P}_{\text{GSP}}(\lambda_m)$
[57]	Photoplethysmogram; Multimodal features	Coronary Artery Disease detection. graph with their correlations as edge weights	Graph-based features: Variations of the top three eigenvalues of the Laplacian matrix. $[\lambda_1 - \lambda_2 \quad \lambda_2 - \lambda_3]$
[55]*	fMRI; Alzheimer's Disease detection.	Correlation, coherence and Granger Causality of functional data across ROIs	Graph-based features: Signal projected onto the high graph frequency band. $\mathbf{f} = \mathbf{U}_h^T \mathbf{x}$
[59]	fMRI; Participants' rates regression.	Connectivity of fMRI data, geometric graph network	Graph-based features: Functional signals filtered by the Warped Translates (WT) generated from functional connectivity. $\mathbf{y}_{\text{rates}} = \theta \cdot \text{WT}(\mathbf{x})$
[60], [61]	fMRI; Cognitive learning task.	Functional data coherence across ROIs	Graph-based features: Total variations (TV): the graph signal smoothness. $\text{TV} = \frac{\sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} w_{ij} (x_i - y_i)^2 \right)}{\sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} w_{ij} \right)}$

$\mathbf{x}$ is a graph signal; x_i and x_j represent signals residing on node i and j , respectively; λ_m and $\mathbf{U}(\lambda_m)$ for $m \in M$ are the i th eigenvalue and eigenvector of a graph Laplacian matrix; M is the number of network nodes; $\mathbf{U}_l$ and $\mathbf{U}_h$ are the low and high graph frequencies, respectively; λ_1, λ_2 and λ_3 are the top-three eigenvalues; $\mathbf{P}_{\text{GSP}}$ is the power in the graph domain; w_{ij} represents the weights of the edges; $\mathbf{P}$ is the whitened matrix; $\mathbf{B}_{\lambda_0}$ and $\mathbf{B}_{\lambda_{\max}}$ are eigenvectors corresponding to the smallest and largest eigenvalues of the whitened covariance matrix; $\mathbf{x}_t$ is a graph signal at time t ; $\mathbf{U}$ is a graph eigenvector matrix; $\hat{\mathbf{X}}_t$ is the normalized signal in the graph space; $\mathbf{P}_{\text{SFM}}$ is the projection matrix for the diagonalization; T is the signal length; θ is the linear regression parameter vector; $\mathbf{y}_{\text{rates}}$ is the participants' rates vector.

*: Three different graph shift operators were used in this study, including the Laplacian matrix, the Jordan decomposition of Adjacency matrix, and the Graph optimisation-based shift operator. Besides [55], the remaining studies in this table used Laplacian matrices as the graph shift operator.

signals onto the eigenvectors corresponding to the largest and the smallest eigenvalues can provide feature vectors to distinguish two groups of EEG signals [51]. A similar diagram [58] has been proposed to generate shared graph eigenvectors of pre-whitened covariance matrix calculated from graph-domain signals for two groups: ASD and neurotypical subjects. The signal in the graph domain was generated using GFT. The shared eigenvectors were obtained to simultaneously diagonalize whitened data covariance of two groups, which can potentially project graph-space signals to discriminative subspaces. This extension is recognized as the Spatial Filtering Method (SFM). According to classification performance, the proposed GSP and SFM integrated method outperformed individual GSP or SFM approach regarding accuracy.

The application of GSP in the BCI settings has been studied in various recording modalities. Moreover, it has been applied to a broad range of neuroscience studies to identify different stages of brain injury from scalp EEG recordings [56], detect epileptic seizures from iEEG [49], and classify fMRI function connectivity networks of participants of different ages [54].

Instead of constructing a graph network across multiple EEG channels or fMRI recordings across regions of interest (ROI), the visibility graph method was applied to convert a time series to a connectivity network, on which an eigenmap can be produced [63]. The averaged power spectrum density in the graph domain was computed from eigenmaps generated from the visibility graph, and showed to be higher for the epilepsy seizures than the healthy control recordings [52].

Diverse scenarios of biological research involve graph networks of multimodal data. The multi-relational graph network was constructed from photoplethysmogram (PPG) and metadata (such as age, weight, height, and gender) for each patient. Specifically, subgraphs of PPG signal features were connected to a mediated graph generated from metadata with the fully-connecting setting. The differences among the largest three eigenvalues were calculated as features to classify patients with coronary artery disease and healthy controls [57]. The classification performance of the proposed method outperformed the state of the art methods regarding accuracy [64], [65].

Three sets of graph transform basis can be generated as introduced in Section III-B and III-E, including the eigenvalue of the Laplacian matrix, the Jordan decomposition of the adjacency matrix, and the graph basis that can optimize the data clustering [55]. The same graph neural network method was applied onto different graph basis to detect Alzheimer's disease from brain functional connectivity constructed from the resting-state fMRI recordings. The results indicated that the clustering-based graph basis achieved the optimal classification performance, which implies the graph signal decomposition for the optimization of specific objectives can help with particular tasks.

Although both the adjacency matrix and Laplacian matrix can describe the graph network connection, the eigenmaps generated from them can differ in terms of computation and effectiveness for specific tasks. Nguyen and Wei [66] developed three graph matrix: adjacency matrix, Laplacian matrix, and the pseudo inverse of the Laplacian matrix from the geometric connection network for different chemicals of the protein-ligand binding to

predict biomolecular macroscopic properties. The same machine learning approach was applied to three types of graph matrix, and the results implied that the adjacency matrix achieved the optimal performance. This result might be inspiring for future research as the adjacency matrix is one of the simplest methods that require less computational power but can still capture the interactions between protein and ligand atoms.

Study [59] investigated the feature extracted using graph wavelet transform. The graph wavelet method used was Warped Translates, which can refine signal representations in graph domain according to eigenvalues of specific network as demonstrated in Section III-D1. The Warped Translates incorporated from fMRI connectivity network were applied to fMRI data to predict the subject's ratings on emotional content in pictures, which showed improved repression performance compared with results without translate or with spline translate.

3) Interpretation of Cognitive Process or Neurological Disorder: Structural pathways are wiring brain regions, and further modulate brain functional activities. The modulation pattern between functional data and the underlying structural network has been discovered to be related to cognitive processes and brain organisations [67]. The low-frequency graph component indicates coupling between brain functional and structural modules, while the high-frequency component implies decoupling modulation. The coupling pattern was quantified by the structural-decoupling index, which was calculated as the ratio of graph signals projected in the high- and low-frequency space. With the Human Connectome Project data, this study discovered that the coupled structural-functional pattern corresponds to brain lower-level functions, such as sensory and motor, while the decoupled pattern relates to more complex functions, such as memory and emotion.

A similar diagram was applied to study a specific cognitive mechanism. It was analyzed the brain fMRI networks, constructed using coherence matrix during a learning task that participants responded to sequentially presented stimuli using a customized response box [60], [61]. The low-frequency graph and the high-frequency graph were shown to be more spatially extended than the bandpass frequency graph, which may imply that the brain is dominated by regularity (low-frequency graph) and disorder (high-frequency graph) during learning. Moreover, the learning rate was positively correlated with the low-frequency graph (smoother) and the high-frequency graph (less smooth) at the beginning and end of the learning process, respectively. The interpretation of this finding is that the cognitive process may start by grasping the big picture of the task at hand, then refine the smaller details to increase the performance on that particular cognitive task.

The vertex-frequency method, the counterpart of the time-frequency analysis in the graph domain, has been proposed to study the EEG network changes between consecutive swallows with numerous liquids in patients with dysphagia [68], which is a swallowing disorder that can be caused by neurological disorders. The energy change among consecutive swallows was positively correlated with the thickness of the liquid, which indicated that dysphagia due to neurological disorder might be caused by muscular forced fatigue.

Moreover, the spatial graph variation was discovered to be related to the high temporal variation in an epicardial electrogram study [69]. A joint graph and short-time Fourier transform method were proposed to analyze epicardial electrogram recordings in the graph, time, and frequency domains. This method first performed a short-time Fourier transform. The frequency bandpass filtered signal in a time interval was further filtered by the graph eigenmodes, which was generated from the eigendecomposition of the graph network. The graph network in this study was with array electrodes as vertices, and each vertex was connected to its nearest eight neighbors with edges measured as the inverse of their distances. Based on the assumption that the ventricular activity is smoother over the graph than the atrial activity, the ventricular activity was estimated as a smooth graph reconstruction problem.

IV. GRAPH NEURAL NETWORK

The graph network is a natural representation of data, which has been massively accumulated in biological research with the emergence of improved data collection techniques. In omics, it has been shown the intra- and inter-network interactions of biological molecules and their associations with diseases and drug treatment [12]. In neuroscience, the network graph has been used to study brain functional and structural connectivity and their associations with various neurological disorders and cognitive states [11].

Deep learning that stacks multiple neural layers has achieved success in various tasks, especially in imaging analysis and computer vision. The critical factor is its ability to aggregate and pass local statistics of structured data sampled on the underlying Euclidean space grid. There has been an increasing interest in applying deep learning to the non-Euclidean structured graph data. In Section IV-B, we provide mathematic frameworks of GNN variants using graph filtering formulated in Section IV-A.

The applications of GNNs for biological networks are reviewed in Section IV-C, based on different levels of network targets: node-, edges- (interaction), and graph-wise classification or prediction for specific biological objectives. Particularly, we summarize different GNN frameworks applied to multi-relational biological networks, which are prevalent in biological research.

A. Convolutional Operation of GNN

The information aggregation of a graph signal $\mathbf{x}$ across a network can be implemented using a graph filter via graph convolution operation as shown in (3). The graph neural network (GNN) can be established by cascading multiple layers of graph aggregation operator with nonlinear transforms introduced. Under the graph filtering framework [22], the l th layer of GNN can be derived from (3) as

$$\mathbf{x}^{l+1} = \rho(H(\mathbf{L})\mathbf{x}^l\mathbf{W}^l), \quad (15)$$

where $\mathbf{x}^l$ gives the input graph signal or features to the l th layer with dimension $M \times N^{l-1}$, $H(\mathbf{L})$ is the graph filter, and $\mathbf{W}^l$ provides the learnable weights of the l th layer with dimension $N^{l-1} \times N^l$. $\rho(\cdot)$ is the nonlinear activation function. One of

the major applications of GSP to biological data is to enhance fMRI data using diffusion MRI via graph filter, which now can be readily implemented using GNNs with this filter-centered framework in (15).

B. Variants of GNN

In this section, we introduce major variants of GNN that have been applied for biological objectives. We develop formulations of various GNNs under the convolutional operator of graph filtering stemming from (3). We integrate variants of GNN that were developed with graph spectral analysis, spatial aggregation, and diffusion processes under a uniform framework - graph filtering of GSP.

Spectral GNN. A spectral GNN can be directly derived from (15) by performing graph spectral decomposition [70]

$$\mathbf{x}^{l+1} = \rho(\mathbf{U}H(\mathbf{\Lambda})\mathbf{U}^T\mathbf{x}^l \cdot \mathbf{W}^l), \quad (16)$$

where $\mathbf{x}^{l+1}$ and $\mathbf{x}^l$ are latent node outputs of layers $l+1$ and l , respectively; $H(\mathbf{\Lambda})$ is the function of the eigenvalue matrix $\mathbf{\Lambda}$. Similar to (1), $\mathbf{\Lambda}$ and $\mathbf{U}$ are the eigenvalues and eigenvectors computed from the spectral decomposition of the normalized Laplacian matrix. $\mathbf{W}^l$ provides the learnable weights of the l th layer. However, there are two significant limitations of the spectral GNN. First, the convolutional operation in (16) is computationally demanding because of the multiplication with the eigenvector and the eigendecomposition of the Laplacian matrix, especially for large graphs. Second, the eigenmode of the Laplacian matrix may not be localized, which is crucial for the extraction of local stationary statistics.

ChebNet. To mitigate the computational and localized issues of the spectral GNN, Defferrard *et al.* [71] proposed that graph filter $g(\mathbf{\Lambda})$ can be well approximated with truncated Chebyshev Polynomials $T_k(x)$ as

$$g(\mathbf{\Lambda}) \approx \sum_{k=0}^K \theta'_k T_k(\tilde{\mathbf{\Lambda}}), \quad (17)$$

where $\tilde{\mathbf{\Lambda}}$ is $\frac{2}{\lambda_{\max}}\mathbf{\Lambda} - \mathbf{I}$. $\lambda_{\max}$ is the maximum eigenvalue of $\mathbf{L}$. $\mathbf{I}$ is the identity matrix. K and θ'_k represent the K th order and coefficients of the Chebyshev Polynomial, respectively. The Chebyshev polynomial can be computed efficiently with the iterations defined as $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$ with $T_0(x) = 1$ and $T_1(x) = x$. We can substitute $g(\mathbf{\Lambda})$ in (16) with (17), and then obtain the updated l th convolutional layer as

$$\mathbf{x}^{l+1} = \rho\left(\sum_{k=0}^K \theta'_k T_k(\tilde{\mathbf{L}})\mathbf{x}^l\right), \quad (18)$$

where $\tilde{\mathbf{L}} = \frac{2}{\lambda_{\max}}\mathbf{L} - \mathbf{I}$. This can be derived with $(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T)^k = \mathbf{U}\mathbf{\Lambda}^k\mathbf{U}^T$.

GCN. With the truncation of the polynomial in (18) up to the K th order, the aggregation is localized at K hops away from the central node. Because $B_K = A + A^2 + \dots + A^K$ gives the K -neighbourhood of a node, where A is the adjacency matrix of a graph. This K -neighbour property also holds when B_k is the summation of the powers of the Laplacian matrix [72]. Kipf and

Welling [73] showed that the truncation of the polynomial with one hop in one layer could alleviate the overfitting problem, while the K -neighbourhood localization can be achieved by multiple layers of convolutional networks. This is one of the most commonly used GNN propagation rules that can be modeled as

$$\mathbf{x}^{l+1} = \rho \left(\tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \cdot \mathbf{x}^l \cdot \mathbf{W}^l \right), \quad (19)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix added with self-loops and $\tilde{\mathbf{D}}$ is the corresponding degree matrix. $\mathbf{W}^l$ provides the learnable weights of the l th layer. $\mathbf{x}^l$ and $\mathbf{x}^{l+1}$ are the outputs of the l th and $(l + 1)$ th layers, respectively.

GraphSAGE. An inductive variant of GNN [74] is proposed to update embeddings for each node using

$$\mathbf{x}^{l+1}(n) = \rho \left(\mathbf{W}^{l+1} \cdot \{ \text{MEAN}[\mathbf{x}^l(i), i \in N(n)] \parallel \mathbf{x}^l(n) \} \right) \quad (20)$$

where $N(n)$ is the neighbourhood of vertex n . This mean aggregator is closely related to the convolutional operation of graph convolutional network (GCN) in (19), because (20) is a linear approximation of a local spectral convolution. In addition to the mean-based aggregator in (20), the local aggregation can be performed using long short-term memory (LSTM) and a pooling operator to obtain larger expressive capability.

GAT. Instead of aggregating local nodes without discrimination, graph attention networks (GAT) [75] introduced an attention mechanism to assign importance weights to nodes. The aggregating computation of GAT can be readily developed from (15) with graph filter $H(\mathbf{L})$ being substituted by an edge-varying filter Φ [76]. The l th layer operation of GAT can be formulated as

$$\mathbf{h}^{l+1} = \rho(\Phi^l \mathbf{x}^l \mathbf{W}^l), \quad (21)$$

where Φ^l is layer-dependent and its element ϕ_{ij}^l represents the importance of node i to node j that can be updated by

$$\phi_{ij}^l = \frac{\exp(\alpha_{ij})}{\sum_{j' \in N(i)} \exp(\alpha_{ij'})}, \quad (22)$$

where α_{ij} is the edge score and

$$\alpha_{ij} = \rho \left(\mathbf{e}^T \left[\mathbf{x}_{l-1} \mathbf{B}_i, \mathbf{x}_{l-1} \mathbf{B}_j \right]^T \right). \quad (23)$$

The matrix $\mathbf{B}$ is introduced with dimension $N^{l-1} \times N^l$. $[\mathbf{x}_{l-1} \mathbf{B}]_i$ is the i th row of the matrix multiplication representing the updated feature of node i , and $\mathbf{e}$ is a vector $2 N^{l-1} \times 1$.

DCNN. Instead of stacking multiple convolutional layers, the diffusion convolution neural network (DCNN) has been proposed with a single layer. The DCNN includes a set of diffusion processes regarding various depths [77]. The convolutional computation of DCNN is similar to (15) but the graph filter is the diffusion process $\mathbf{L}^f$, which is the f th order term in polynomial form of $H(\mathbf{L})$. Moreover, the learnable weights in (15) are performed to compute weighted sum among different node features, while the element-wise weights are learned in DCNN without combination across features. The diffusion process of DCNN with the f th depth can be formulated as

$$\mathbf{h}^f = \rho(\mathbf{w}^f \odot \mathbf{L}^f \mathbf{x}), \quad (24)$$

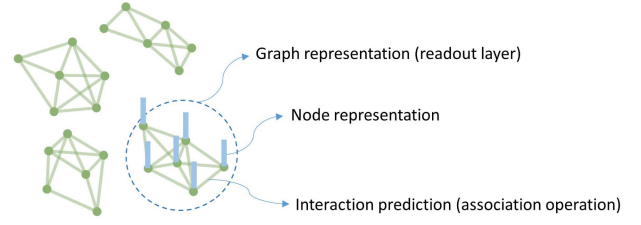


Fig. 5. Different levels of GNN learning: node, edge, and graph. GNNs can be applied to perform node-wise, edge-wise (interaction prediction), and graph-wise learning for different biological data applications.

where $\mathbf{h}^f$ is the hidden state of the f th depth or layer, $\mathbf{w}^f$ is the learnable weights of the f th layer, while $\mathbf{L}^f$ is the f th power of the Laplacian matrix, which is the f th order term in the polynomial form of $H(\mathbf{L})$. $\odot$ represents element-wise multiplication. The output dimension of the f th depth is the same as the dimension of the input graph signal $M \times N$. The output of DCNN is the concatenation of all the diffusion process depths $\{\mathbf{h}^f\}_{f=1}^F$, where F is the number of layers.

C. Applications

GNNs can be applied to perform node-wise, edge-wise (interaction prediction), and graph-wise learning for different biological data applications, as illustrated in Fig. 5. For node-wise learning, the ultimate goal is to generate node representations, and the target of interest is the node component in the graph. Similarly, edge-wise and graph-wise learning represent research of interest are edge component and global structure of graphs, respectively.

The node-wise GNN can be considered as a preliminary step of the graph-wise and edge-wise graph learning for biological networks. Specifically, the graph-wise learning and edge-wise learning apply the readout layer and the association operation, respectively, to the learned node representations that generate graph-level representations and association scores. The reviews of GNNs used to biological data with node, edge, and graph-oriented learning procedures are given in Sections IV-C1, IV-C2 and IV-C3, respectively. In Sections IV-C4 and IV-C5, we review two classes of special GNN applications for multiple biological graphs and medical images, respectively.

1) Node-Wise GNN for Biological Data: Machine learning can be performed with the inductive learning and the transductive learning manners. Inductive learning tries to build a general predictive model from observed instances, which can perform the inference for the unseen instances. In contrast, transductive learning is trained on both labeled and unlabelled samples, and the learned model can assign labels for unlabelled instances. Both learning frameworks are useful in the application of node-wise GNNs for biological data, which can be brain ROIs, cells, and molecules.

The applications of inductive learning and transductive learning for the node-wise classification task on graph networks are illustrated in Fig. 6(a) and (b), respectively. As shown in Fig. 6(a), GNN methods can be applied to the training networks to generate a general predictive model, which can predict node

TABLE IV
NODE-WISE CLASSIFICATIONS USING GNN FOR OMICS DATA

Ref.	Network	GNNs	Loss function	Application
Inductive learning				
[75]	PPI	GAT	Supervised loss: cross-entropy function of the labeled nodes	Labels for gene ontology
[79]	PPI	Multi-hop hierarchical GNN	Supervised loss: cross-entropy function of the labeled nodes	Labels for gene ontology
[80]	Cell graphs	GAT	Supervised loss: negative log-likelihood loss	Classification of healthy and multiple sclerosis
Transductive learning				
[81]	Gene-gene interaction	GCN	Supervised loss: cross-entropy function of the labeled nodes	Cancer drive genes
[82]	PPI	GCN	Supervised loss: labeled nodes Unsupervised loss: similarity of neighboring unlabelled nodes	Protein complexes detection
[83]	PPI	GraphSAGE	Supervised loss: cross-entropy function of the labeled nodes Unsupervised loss: penalty of the dissimilarity of neighboring nodes	Autism-gene association
[84]	Multi-relational PPI	GCN	Supervised loss: cross-entropy function of the labeled nodes Unsupervised loss: label smooth estimation and the sparsity of the mixing matrix	Protein function prediction
[85]	PCG-PCG interaction, PCG-miRNA interaction	GCN	Supervised loss: cross-entropy function of the labeled nodes Unsupervised loss: graph Laplacian regularization	MiRNA-disease association

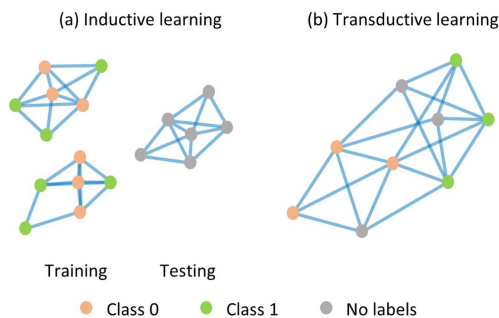


Fig. 6. Inductive learning and transductive learning for node-wise classification. Inductive learning tries to learn a general model from available instances, which can perform inference for the unseen instances. In contrast, transductive learning is trained on both labeled and unlabeled samples, and the learned model can assign labels for unlabelled instances.

labels for unseen networks. As seen in Fig. 6(b), GNNs can also be applied with the transductive learning method to a network, which is composed of labelled and unlabelled nodes. The learned model can predict labels for the unlabelled graph nodes. This section reviews studies that applied GNNs to perform the node-wise classification for biological networks with inductive learning and transductive learning manners.

The node-wise GNNs have been mainly applied to two types of biological networks: omics network and population graph. The review for the omics networks first covers the inductive learning manner, then summarizes the transductive learning approach for the label propagation across nodes. Compared with inductive learning, transductive learning is more commonly used for biological networks. Conventionally, labels for a subset

of network nodes are annotated by the expensive and time-consuming wet-lab experiments. Transductive learning using GNNs is a computational approach that can potentially propagate labels for the remaining nodes. Besides, we review the application of node-wise GNNs to a special biological network - population graphs with patients as network nodes.

a) Omics Networks: The node-wise application of GNNs on omics data or networks is summarized in Table IV. The loss function is given in the fourth column, which can be composed of the supervised term imposed on the labeled nodes and the unsupervised term for specific objectives. This table is categorized as inductive learning and transductive learning as illustrated in Fig. 6.

Veličković *et al.* proposed the GAT, a modified spatial GNN that can assign different weights to nodes in a neighborhood [75]. They tested GAT on the protein-protein interaction (PPI) network with the inductive learning way and achieved the accuracy of 97.3%, which is much higher than the accuracy of 76.8% achieved by GraphSAGE without node weights. Zhang *et al.* [78] applied an extensive number of GCN layers to PPI for the protein family prediction, and the results implied that the increase of GCN layers did not improve performance. Intrinsically, adding one layer of GCN introduces more smoothness between two classes of data. Thus, the shallow architecture of GNN is commonly used, which can capture local information well, but with the limited ability to aggregate further information. A hierarchical GAT architecture was proposed to include multiple hops of aggregations in one layer, on which the attention mechanism as shown in (22) was performed to calculate the importance factor of each hop to the current node [79]. The method was tested on the PPI dataset, and the accuracy was 98.8%, which was slightly better than GAT. GAT was also

applied to cell graph networks composed of gene expression to detect Multiple Sclerosis patients [80], which also showed much better accuracy (92.3%) than the GCN method proposed by Kipf and Welling [73] (72.1%).

Schulte-Sasse *et al.* [81] applied GCN to the gene-gene interaction network with multimodal omics data embedded in the node feature to predict the cancer drive gene. The loss function was the cross-entropy function for the labeled nodes. Moreover, both supervised term for the labeled nodes and unsupervised term for the penalty of the dissimilarity of the neighboring nodes were included in the loss function to predict protein complexes [82] and autism-related genes [83] in the PPI networks using GNNs.

The label propagation is performed across multi-relational interactions that are available for biological targets. Multiple PPI networks can be generated across various cells, and the nodes of proteins can be associated with functions. Ioannidis *et al.* [84] applied GCN to each relational PPI, and the learned node representations across multiple relational graphs were combined with a learnable mixing matrix. Besides the cross-entropy for the labeled nodes, the loss function included the regularization of the sparsity of the mixing matrix and the smooth label estimation.

The label propagation with the transductive learning of GNNs cannot only be performed within a single network. In some cases, the labels for one set of biological targets are available, and we can propagate labels for another set of molecules across the bridging interaction networks between two groups of molecules. MicroRNA plays a critical role in the biological processes of diseases. The disease labels for protein-coding genes (PCG) have been well investigated, and the PCG-PCG interaction and PCG-miRNA interactions are available. GCN can be applied to propagate PCG disease labels across the integrated PCG-PCG and PCG-miRNA interactions to miRNA [85]. The loss function was composed of the supervised term of the labeled nodes and the graph Laplacian regularization that enforced the smoothness of the learned representations across the graph network.

b) Population graph: Diagnostic biomarkers for the patient population with neurological diseases, such as Alzheimer's disease and Autism spectrum disorder, are essential for the research and the following therapy. Graph networks provide a natural tool to analyze interactions in the patient population. In addition to patient interactions regarding imaging data, the non-imaging data such as gender and sites can decode more information in the population graph, which can deliver better biomarkers through multimodal data. In this section, we summarize studies that performed GNNs on the population graph constructed from imaging or medical results along with metadata. We outline the architectures of GNN frameworks that were performed on the multimodal population graph in Fig. 7.

In Fig. 7(a), the node features of the population graph were imaging data, while the edge weights were evaluated by both imaging data and metadata. The imaging data used for the population included MRI imaging morphological measures of different anatomical atlases [86], fMRI data [87], [88], and functional connectivity networks [89]. The metadata used were gender, sites, and equipment [86]–[89]. GNNs were applied to

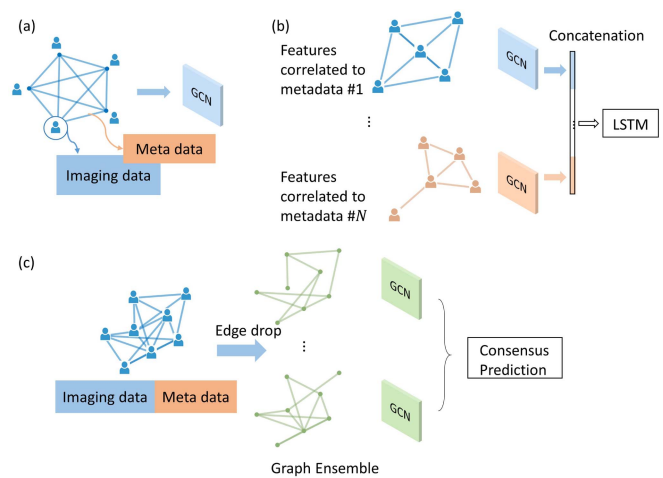


Fig. 7. Application of GNNs on population graphs with multimodal data. (a) The graph was constructed with imaging data and metadata as node features and edge weights. (b) The multiple graphs were generated by metadata and their correlated features. GCN was applied onto each graph, and the learned embeddings were concatenated for the following prediction procedure using long short-term memory [89]. (c) The graph ensemble was generated by the random edge drops from the multimodal population graph. The prediction tasks were performed by the consensus or the average across all the GCNs in the ensemble [90].

learn useful embeddings for mild cognitive impairment (MCI) detection in [86]–[88]. Studies [87], [88] showed that the population graph with metadata outperformed the one without metadata in the MCI classification tasks. In [89], GCN was modified by adding the localization kernel function that takes into account all the neighbors within specific hops. This study showed that the modified GCN with localization kernel outperformed GCN in the task of MCI detection.

Imaging data or medical results used as node features can be independent of the metadata used for the calculation of the network edges. Valenchon *et al.* [90] developed a framework that determined the correlated node features with metadata or their combinations. Thus, multiple graph networks can be generated from the metadata and their correlated features. GCN was applied to each graph, and the learned embeddings were concatenated for the following tasks. The diagram of this method is shown in Fig. 7(b) that was applied to predict the development of MCI to the Alzheimer's disease, which significantly outperformed the GCN prediction without the multiple graph generation module.

The graph network topology is critically vital for GNNs. The underlying graph topology of the population graph is not available, which can potentially deteriorate the performance of GNNs when the inferred graph topology deviates from the ground truth. Anirudh and Thiagarajan [91] proposed a graph ensemble GCN method that can increase the robustness of the performance of the GCN when the graph topology is not available. The framework of the graph ensemble method is demonstrated in Fig. 7(c) that multiple graphs were generated by randomly dropping edges from the population graph, on which GCN was performed as a weaker learner, and the averaged outputs across all the graphs were considered for the following task. They applied this method

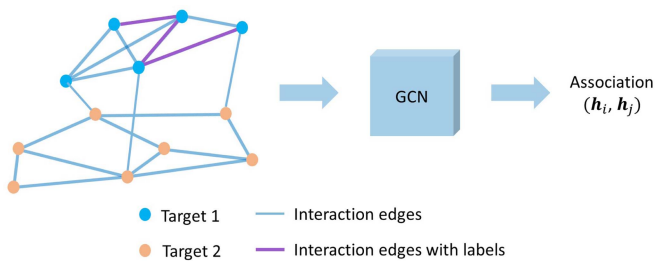


Fig. 8. Interaction prediction framework. This framework can propagate edge labels during transductive learning.

to classify ASD patients and health controls. It has been shown in this work that the ensemble GCN outperformed GCN in this classification task slightly.

Besides the imaging and metadata, the population graph was constructed from patients' omics data to predict cancer survival [92]. The patient graph network was built from two sets of data, mRNA expression and DNA methylation, by updating the k-nearest similarity matrix of each datatype. The min-redundancy and max-relevance algorithm was applied for the feature selection from multiple genomic data and clinical examinations of cancer samples. GCN was used in a semi-supervised manner to predict cancer survival.

2) Edge-Wise GNN for Biological Data: The prediction of network interactions across biological molecules is essential for biological research that can infer protein-protein interactions, gene-disease associations, and drug-target interactions. In this section, we first review the application of the interaction-wise GNNs for the omics networks. Second, we provide a review for the application of GNNs on the drug-target network, which is a type of biological network that involves chemical compounds and plays a critical role in drug discovery.

A widely used biological interaction prediction framework of GNNs is depicted in Fig. 8, which can be categorized as transductive learning that propagates edge labels during the learning procedure. Specifically, node representations can be updated across unimodal or multimodal biological networks using GNNs. Then, the edge or interaction scores are computed by the association operation performed on the learned representations between two nodes.

a) Omics Networks: We summarize studies that performed interaction prediction within a single biological network (such as PPI) and the gene-disease networks in Table V. The edge prediction framework is given in Fig. 8, and the association operations used are listed in the third column of Table V.

The node representations learned from a pair of molecules to be associated can be concatenated and processed by neural networks with the semi-supervised learning manner. The interaction prediction studies based on this concatenation method predicted pairwise association scores for ligand-receptor interface [93], PPI [94], lncRNA-gene interactions [95], and lncRNA-disease interactions [96] using various neural network paradigms, as shown in Table V.

The pair similarity calculated by the inner product was also used to predict network interactions. GCN was modified to

predict synthetic lethality (SL), which is a promising form of gene interaction for cancer therapy with sparse interaction structure [97]. The SL interaction score was predicted using the inner product of the learned embeddings. Also, the edge dropout and node dropout modules were designed to deal with the sparsity of SL interactions. The dual dropout GCN method outperformed the GCN method for the SL prediction task. The inner product was also applied to calculate the association scores for the gene-disease embedding pairs, which were learned from disease similarity and gene similarity networks [98]. Furthermore, Li *et al.* [99] performed the similarity computation for a pair of miRNA and disease with a relation-type-specific disease-miRNA parameter matrix, and the loss function was the summed loss across multiple relational disease-miRNA matrices.

Instead of computing similarities between GNN learned embeddings, the pairwise representations can be first extracted, and GNNs can be performed to predict their association scores. Wang *et al.* [100] proposed a disease and circRNA similarity-based method to predict circRNA-disease associations, which hold the hypothesis that similar circRNAs tend to associate with similar diseases and vice versa. They generated circRNA similarity and disease similarity, which formed the circRNA and disease pairwise fusion data. GCN was applied to the fused pairs to predict new associations.

It is worth noting that gene-disease association studies are covered in both node-wise and edge-wise learning sections, i.e. Sections IV-C1 and IV-C2. Generally, the association performed with the node-wise learning scheme propagates the disease labels for the unlabeled genes during the transductive learning procedure. In contrast, the edge-wise association for gene and disease was computed using association operators performed on the learned node representations.

Drug-Target Interaction prediction. The prediction of drug-target interactions (DTIs) can boost the drug discovery process, which has been traditionally conducted by time-consuming and expensive lab experiments. GNNs provide us a way to predict DTIs from various omics data interaction networks efficiently. Similar to the interaction prediction for omics networks, the framework illustrated in Fig. 8 was applied for the DTIs prediction.

GNNs can be performed on heterogeneous networks, and node representations can be learned with loss function imposed on a subset of edges with labels. In the end, edges without labels can be predicted with the form of $\mathbf{h}_i \mathbf{M}_r \mathbf{h}_j^T$, where $\mathbf{h}_i$ and $\mathbf{h}_j$ are learned representations of nodes i and j , and $\mathbf{M}_r$ is also a learnable matrix that represents the interaction between $\mathbf{h}_i$ and $\mathbf{h}_j$ over a relation r . Zitnik *et al.* [101] applied this framework to the heterogeneous graph network composed of protein-protein interactions, drug-drug interactions, and drug-protein interactions to infer side effects of drug combinations. Likely, Manoochchri *et al.* [102] applied this framework on a broader range of graph networks, including seven interaction networks to predict the drug-protein interactions.

DTIs prediction involves the association computation between two components: proteins and chemical compounds. In addition to the framework shown in Fig. 8, the learning diagram shown in Fig. 9 is used for DTIs prediction that representations

TABLE V
INTERACTION PREDICTION USING GNNs FOR BIOLOGICAL NETWORKS

Ref.	Network	Association	Application
[93]	Structural graph of ligand and receptor proteins	Fully connected and softmax layer	$\text{softmax}(\mathbf{W} \cdot [\mathbf{h}_i, \mathbf{h}_j] + \mathbf{b})$ Protein interface prediction
[94]	PPI	Deep neural network (DNN)	$\text{DNN}(\mathbf{h}_i, \mathbf{h}_j)$ Protein interface prediction
[95]	MiRNA mediated lncRNA-gene network	CNN	$\text{CNN}([\mathbf{h}_i; \mathbf{h}_j])$ LncRNA-gene association
[96]	A heterogeneous network of lncRNA, disease, miRNA	Fully connected and softmax layer	$\text{softmax}(\mathbf{W} \cdot [\mathbf{h}_i, \mathbf{h}_j] + \mathbf{b})$ lncRNA-disease association
[97]	SL graph	Inner product decoder	$\frac{1}{1 + \exp(\mathbf{h}_i \cdot \mathbf{h}_j^T)}$ SL prediction
[98]	Disease similarity network Gene similarity network	Clip of the inner product	$\text{clip}(\mathbf{h}_i \cdot \mathbf{h}_j^T)$ Gene-disease association
[99]	PPI Disease-gene network miRNA-gene network miRNA-disease network	Factorized operation	$\mathbf{h}_i \mathbf{M}_r \mathbf{h}_j^T$ MiRNA-disease association
[100]	Disease semantic similarity network ($G1$) CircRNA similarity network ($G2$) CircRNA-disease adjacency matrix ($G3$)	The fusion of disease similarity and circRNA similarity	Combined disease similarity: $C_d \leftarrow (G1, G3)$ Combined circRNA similarity: $C_r \leftarrow (G2, G3)$ The fusion of disease and circRNA: $[C_d, C_r]$ CircRNA-disease association

$\mathbf{h}_i$ and $\mathbf{h}_j$ are embeddings learned for biological targets to be associated using GNNs; $\text{softmax}(\cdot)$ performs exponential function to each element of the input vector, which is then normalized to ensure the element sum of the output vector equals to one; $\text{clip}(\cdot)$ is the clip function that enforces the value in the range $[0, 1]$; $\mathbf{M}_r$ is the relation-type-specific disease-miRNA parameter matrix;

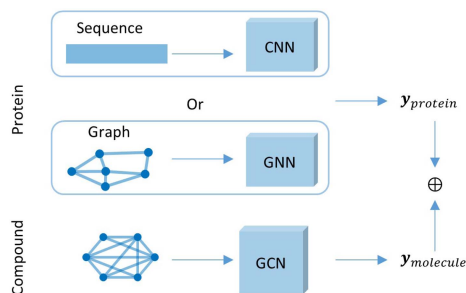


Fig. 9. Interaction prediction framework for DTIs. In this framework, representations for proteins and chemical compounds are learned separately, and their representations can be concatenated for their association prediction.

for proteins and chemical compounds are learned separately, and their representations can be concatenated for their association prediction. Protein representations can be learned from amino acid sequences using CNNs [103]–[105], or more sophisticatedly, they can be learned from the binding pocket with structural information using GNNs [106]. The drug compound can be modeled with the molecular graph, and its embeddings can be learned with GNNs. The concatenation of the protein and drug representations can be used to predict protein-compound interactions. This framework has also been used to annotate

proteins with gene ontology (GO) [107]. The protein representations were learned from the amino acid sequence using CNNs, while the GO was organized with a graph network, and its representation was generated using GCN. Their association was estimated with the inner-product of their representations.

DTIs prediction was also performed using GNNs with the graph-wise classification manner. Lim *et al.* [108] directly extracted graph features from the protein-ligand binding pose in order to learn whether the binding is active or inactive. Furthermore, the manual extraction of DTIs information from the increasing volume of biomedical literature can be time-consuming, which has been investigated using the long short-term memory (LSTM) and GNN computationally [109]. The LSTM was applied to capture the contextual information, and the dependency tree was generated from a sentence. GCN was applied to the dependency tree graph and the contextual knowledge to predict the chemical-protein relations from the literature.

3) Graph-Wise GNN for Biological Data: Graph representations can be generated from node representations learned by GNNs for graph-oriented objectives. We summarize studies that used GNNs to learn biological graph representations in Table VI. The column ‘Readout layer’ indicates the methods used to produce graph embeddings from learned node embeddings.

A straightforward way is to combine learned node embeddings with a permutation invariant function pooling layer, such

TABLE VI
SUMMARY OF GRAPH REPRESENTATION LEARNING

Ref.	Recording modalities	Readout layer	Application
[110]	Omics	Pooling, summation	Bioinformatics network
[111]	MRI	Pooling, average	Detection of AD and MCI
[112]	Omics	Pooling, max	Classification of cancer subtype
[113]	Omics	Pooling, adaptive	Bioinformatics network
[114]	fMRI	Pooling, adaptive	ASD detections
[115]	MRI	Pooling, adaptive	Parkinson's disease score prediction
[116]	MEG	Concatenation, stacked autoencoder	Visual stimuli classification
[117]	EEG	Concatenation, broad learning system	Emotion classification
[118]	fMRI	Concatenation, similarity	Detection of ASD and ADHD

as summation and average functions. Atamna *et al.* [110] applied GCN to the omics network dataset for the graph-wise classification using the summation function to combine learned node embeddings. They introduced the expressiveness item to the cross-entropy loss function, since the expressiveness quantifies the ability of GNNs to learn different embeddings for graphs with different structures. However, their results showed that the highest expressiveness value did not correspond to the optimal classification performance in most cases. Wee *et al.* [111] produced graph representations for the cortical thickness networks to classify AD and MCI patients with the average-pooling readout layer. Moreover, some complex biological processes can involve multiple network interactions. Lee *et al.* [112] investigated the differences among cancer subtypes related to gene pathways that each pathway was modeled as a graph network with gene expression profiles. GCN was applied to learn node representations, on which the max-pooling was used to generate graph-level pathway representations.

The pooling layer using permutation invariant functions are useful and feasible to generate graph representations with the same dimensionality as node representations. However, the underlying graph architecture is not considered in the learned graph embeddings. Xie *et al.* [113] applied GNN to learn node embeddings and DIFFPOOL to learn graph embeddings, on which the node-wise classification and graph-wise classification were performed. This framework made use of node labels to improve the performance of graph classification. The DIFFPOOL module was applied to reduce the dimensionality of node representations adapted to the graph clusters, on which the permutation invariant function summation generated the graph representations. Moreover, adaptive pooling modules were introduced to compute graph representations for the ASD patient detection task using fMRI connectivity networks [114], and for the state prediction of Parkinson's disease using MRI structural networks [115].

Studies also kept all the learned node embeddings for the following classification task and can contribute to the feature

space with a high dimensionality, challenging the following tasks. Guo *et al.* [116] concatenated all the nodes representations to classify MEG functional connectivity network with different visual stimuli. They introduced a stacked autoencoder to reduce the dimensionality of the learned representations. Wang *et al.* [117] learned the graph representations from EEG recordings using ChebNet with multiple orders of Chebyshev polynomial approximation. The learned graph representations across all the Chebyshev polynomial orders were processed by a broad learning mode using a flat network to learn discriminative graph features, for an emotional state classification task. Results indicated that this diagram showed excellent emotion recognition performance.

The reduced dimensionality of the graph representations is not always required. Yao *et al.* [118] developed a framework to learn high-dimensional graph representations for the candidate patient using negative group patients, and positive group patients synchronously. The similarities of the candidate patient with the positive group and the negative group were calculated for the classification.

Furthermore, graph-wise classification can be converted to node-wise classification for hierarchically structured graphs that nodes are subgraphs connected by a high-level graph. The subgraph representations are the node features of the high-level graph, which can be further updated by GNN operations. However, variant sizes of the graph representations can be generated from subgraphs with variant numbers of nodes. Li *et al.* [119] proposed to use the attention mechanism to unify the size of graph representations according to the importance of nodes in subgraphs. They tested this hierarchical GNN method on the PROTEINS dataset, and the accuracy was 77.26%, which outperformed the graph classification method that with expressiveness imposed in the loss function [110] and the node label supervised learning GNN method [113] listed in Table VI with accuracies of 72.06% and 76.18%, respectively. The comparison may imply that the hierarchical graph learning structure and node label supervised learning can take advantage of graph structures and more labels, which could boost the classification performance.

4) GNN of Multiple Graphs: It is quite often to have multiple relational graph networks in biological research. The available biological data can be various medical examination results of patients, brain imaging data across a group of participants and molecular interactions across multiple omics data for purposes of disease diagnosis and treatment, drug discovery, and pathological research. In the literature, there were three major GNN frameworks for multiple graphs, including independent GNN, interactive GNN, and hierarchical GNN. Their basic architectures are shown in Fig. 10.

The independent GNN framework applied a GNN module to each graph network independently, and the learned graph embeddings from multiple graphs were concatenated for the following tasks. Its basic architecture is shown in Fig. 10(a). This paradigm was used for numerous graph networks constructed by multiple medical examinations and heterogeneous brain networks to predict the development of Alzheimer's disease [90] and to identify mental disorders [115]. Likely, the

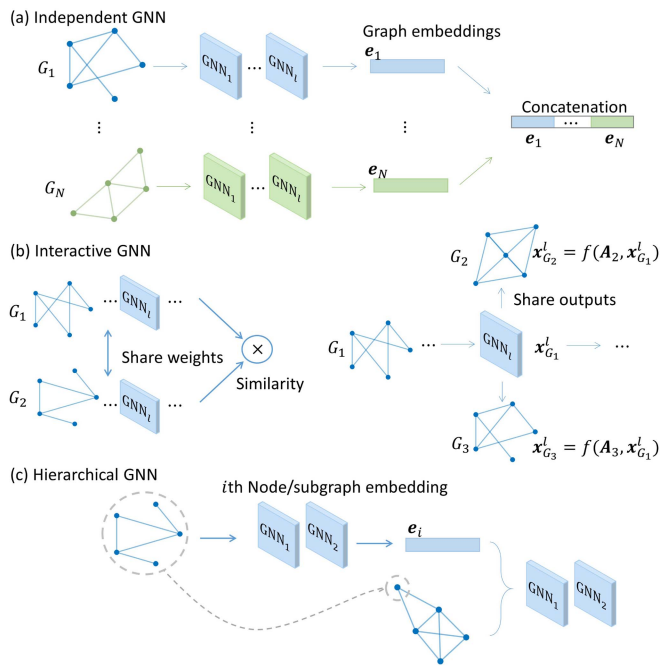


Fig. 10. GNNs across multiple graphs. (a) Independent GNN. GNN was applied to multiple graph networks independently to generate graph embeddings that were concatenated for the following tasks. (b) Interactive GNN. GNN applied to multiple graphs interactively with either their learning weights or layer outputs shared. (c) Hierarchical GNN. Multiple graphs were modeled as nodes of a hierarchical graph, for which GNN was applied onto the node graph and hierarchical graph to learn embeddings for a specific task.

independent GNN was applied to multiple biological pathways to generate their embeddings, which were combined with the attention mechanism for the cancer subtype classification [112]. This framework was modified and applied to learn protein functions that node embeddings of a multi-relational network were learned by GNN independently. Their associations were learnable parameters and were used to combined embeddings across multiple networks [84]. This type of architecture can be useful when there is no clear correlation among various relational graph networks.

The graph aggregation operations can be performed in an interactive manner for biological significance. This paradigm is illustrated in the left panel of Fig. 10(b) that interactions can be achieved by sharing weights or sharing convolutional layer outputs. The sharing weights approach was applied to learn the similarity of brain functional networks among multiple subjects [120]. The sharing weights GNN can guarantee that two brain networks were processed in the same manner, producing biologically significant graph similarity. The learned similarity score can be used for the following classification task that the similarity for samples from the same class was enhanced, and the similarity for samples from different classes was penalized. This framework was also used to learn representations and their similarity scores from multiple brain networks for subjects classification [118].

The interaction that shares the convolutional layer outputs can bridge two graph networks by an intermediate network.

GNN was applied to the intermediate graph network to produce convolutional outputs $x^l_{G_1}$ for each layer that was fed into other two graph networks, which was multiplied with their adjacency matrices A_2 and A_3 . The architecture of this diagram is shown in the right panel of Fig. 10(b). This framework has been applied to predict microRNA-disease association that GNN was applied to PPI as the intermediate network to bridge and update microRNA-gene and disease-gene networks [99]. For the interactive GNN, the sharing weight manner can be beneficial when the similarity between the two graphs is of interest. The sharing outputs diagram can be applied when an intermediate graph can integrate other graphs, and the same processing procedure across multiple graphs is not necessarily required.

Furthermore, interactions across multiple graphs can be established through the hierarchical graph illustrated in Fig. 10(c). Two levels of GNNs were applied to learn embeddings for subgraphs and the hierarchical graph. GNN was first applied to the node level graphs to learn embeddings. Then, GNN can be applied to the hierarchical graph to learn representations for a particular task with the first level of learned graph embeddings as node features. In order to deal with variable node sizes of subgraphs, the self-attention mechanism in [119] was applied to learn the importance score for each node embedding to the subgraph, then node embeddings were integrated with importance scores to generate subgraph representations with invariant size. This architecture can be helpful when this hierarchical structure is biologically plausible. The dataset PROTEINS [121] is a graph network where nodes are secondary structure elements and edges between nodes representing two nodes are neighbors in the amino-acid sequence or in 3D space. The hierarchical GNN framework was validated on the PROTEINS dataset, and obtained 1.27%-5.59% improved accuracy compared with benchmark graph methods.

5) GNN for Medical Images: Medical images play a vital role in diagnosis and treatment, which can reveal alterations in underlying tissues or organs. It is beneficial in clinics to perform automatic image segmentation and abnormality detection on medical images, which can either take experts a plenty of time or hinder diagnosis when experts are not available. Images are regular Euclidean data that can also be modelled as graph networks with following advantages: a) model local configurations of images; b) model sequential structures in images with hierarchical relations, such as arteries that bifurcates; c) require less memory and runtime with only graph nodes and nodes features to be considered.

The key step in the application of GNN for medical images is to convert images to network structured data. Pati *et al.* [122] proposed a GNN diagram to predict cancer grades from hierarchical networks composed of tissue graphs and cell graphs, which can potentially link particular cell configurations lies in heterogeneous tissues. The tissue and cell graphs were generated with biologically plausible assumptions that adjacent tissue or cells should be connected in networks. Jiménez-Sánchez *et al.* [123] generated graph nodes by hierarchical clustering of image pixels, while the edges were associated with the sizes and distances of two vertices. GNN was applied in an unsupervised manner to perform image segmentation, and the learned

representations were shown to be able to classify tissues with different histopathological characteristics.

In addition, particular tissue structures can be modelled as networks. Arteries are sequential data with bifurcations that have been modeled as graph networks whenever there is a bifurcation, a new node is added [124]. The position and image features were extracted for each graph node, and two layers of GCN were constructed for the arteries label prediction task.

CNN has been utilized as a pre-processing step to learn features from medical images, on which networks are constructed for the following GNN aggregation. Yu *et al.* [125] applied this scheme to first learn features using CNN from X-ray images, and then each feature is enhanced via neighbouring aggregation with GNN to classify images into normal and pneumonia. The same diagram was applied by Shin *et al.* [126] that first generate vessel feature and probability map from a coronary artery X-ray image, then graph network is constructed on vessel maps for the local structural learning via GNN. With both CNN global and GNN local embeddings, this method gives the pixel-wise classification result for vessel segmentation.

The encoder-decoder method has been applied to medical images to generate meaningful representations such as medical images and tissue contours. Han *et al.* [127] proposed a symbolic graph embedded encoder-decoder scheme to generate vivid spinal images. A single symbolic graph is constructed from all the images in the training dataset. The CNN encoder outputs were embedded into the symbolic network with the graph attention mechanism, following which GCN was applied to propagate representations on the symbolic graph. A similar attention mechanism was used to embed updated representations of a symbolic graph into the input of decoder, which outputs generative images for adversarial training for segmentation.

A similar CNN encoder-decoder scheme was applied by Hansen *et al.* [128] to predict the displacement of medical images for the image registration task. Considering the efficiency of memory and runtime, the 3D images are transformed into graph networks with keypoints, on which CNN encoder was applied to learn displacement embeddings. The learned embeddings were further aggregated along the keypoints graph using GNN. Then, a CNN decoder was applied to estimate the displacement.

Tian *et al.* [129] applied CNN encoder and convolutional LSTM (ConvLSTM) decoder to 3D slices of MRI data to produce prostate contour for cancer diagnosis. The ConvLSTM was used to delineate vertices on the prostate contour, and all the vertices comprise a graph network. The GNN module was designed to interactively correct locations of vertices on a contour. A manual relocation of a vertex is propagated to its neighbouring vertex with a GNN operation. A corrected contour can be generated with the GNN module.

V. GRAPH NETWORK TOPOLOGIES

The natural graph structure is available for an array of biological data. In omics, PPI and gene-disease association have been discovered by different wet-lab experiments [84], [97], [110], [130], [131]. In neuroscience, the brain structural network can be generated with brain imaging techniques [17]–[19], [41], [47].

However, the graph network is not always available in biology. For instance, the population graph with patients as nodes does not have a natural connectivity edge between patients [91]. Also, the graph network based on three-dimensional protein structure does not hold natural network connectivity [93]. In biology, graph nodes can be legitimately chosen as brain ROIs, genes, diseases, proteins, or patients, while the connection edges should be generated appropriately according to specific applications.

It is straightforward to use the similarity of node signals to infer the graph topology, which have been broadly applied in biological network studies [50], [53], [55]–[132]. Similarity measures that are directly calculated from data recordings can be sensitive to noises. Besides, naïve similarity measures, such as correlation and coherence, do not eliminate indirect connections, so that they tend to overestimate the connection network. The assumption of data models or constraints can help resolve these issues and reveal intrinsic connections in the graph network.

In this section, we cover the model-based graph topology inference methods that introduce assumptions about the node data relations or the graph structure to recover the underlying network interactions. Formulations of the graph learning inference methods are given in Section V-A. In section V-B, we review the applications of graph learning methods in biological data, which mainly lies in the brain connectivity network inference.

A. Graph Learning Inference

The assumption of data models or the additional constraints can help to recover the underlying network structure with less distortion introduced by noises. They also can reduce the over-estimation issue introduced by the similarity measures. One intuitive assumption is the sparsity of the graph that we tend to model the data with a graph network with the least number of connected edges.

Following the notation given in Section III-C, we can write a graph signal matrix $\mathbf{x}$ as

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_M \end{bmatrix}, \quad (25)$$

where the row vector $\mathbf{x}_i$ represents data on the i th vertex. The dependency of vertex i on the remaining vertices with a sparsity constraint can be modeled by the minimization problem as [133]

$$\min_{\beta_i} \left\| \mathbf{x}_i^T - \mathbf{x}_{\setminus i}^T \beta_i \right\|_2^2 + \rho \|\beta_i\|_1, \quad (26)$$

where $\mathbf{x}_{\setminus i}$ is the matrix $\mathbf{x}$ with the i th row removed, β_i is a vector with $M - 1$ dimension, and $\|\cdot\|_1$ is the L_1 norm. The first term in (26) promotes the correlation relationship between i th vertex and the remaining vertices, while the second term imposes the sparsity for their correlation coefficients β_i with the L_1 norm. The minimization problem of (26) can be solved by the well-developed least absolute shrinkage and selection operator (LASSO) [134].

Instead of the linear assumptions imposed on the data relation, we can directly estimate the vertex data from the observed data with the assumption of its smooth distribution on the graph. We can denote the estimated graph data as $\hat{\mathbf{x}}$ and its smoothness across all the observations is $\text{Trace}\{\hat{\mathbf{x}}^T \mathbf{L} \hat{\mathbf{x}}\}$. Besides the smoothness constraint, the energy constraint on the graph Laplacian matrix $\|\mathbf{L}\|_F^2$ can be imposed, which confines the scaling of the graph Laplacian matrix. The data model with both smoothness and energy constraints can be expressed as [135]

$$\min_{\hat{\mathbf{x}}, \mathbf{L}} \frac{1}{2} \|\hat{\mathbf{x}} - \mathbf{x}\|_F^2 + \alpha \text{Trace}\{\hat{\mathbf{x}}^T \mathbf{L} \hat{\mathbf{x}}\} + \beta \|\mathbf{L}\|_F^2, \quad (27)$$

where $\|\cdot\|_F$ is the Frobenius norm of a matrix, α and β are penalty parameters for the smoothness and energy constraints. $\text{Trace}\{\cdot\}$ is the trace operation of a matrix that returns the sum of the diagonal elements.

Another approach is to learn the generalized Laplacian matrix $\mathbf{Q}$ of the graph with the smoothness assumption directly from data [136]. The generalized Laplacian matrix is the Laplacian matrix with self-loops added $\mathbf{Q} = \mathbf{L} + \mathbf{P}$, where $\mathbf{P}$ is a diagonal matrix with unit entities for vertices with self-loops. The smoothness optimization of the graph signal on the generalized Laplacian matrix is

$$J = \frac{1}{N} \text{Trace}\{\mathbf{x}^T \mathbf{Q} \mathbf{x}\} = \text{Trace}\{\mathbf{x}^T \mathbf{x} \mathbf{Q}\} = \text{Trace}\{\mathbf{R} \mathbf{Q}\}, \quad (28)$$

where $\mathbf{R}$ is the covariance matrix of the graph data $\mathbf{x}$. The generalized Laplacian matrix $\mathbf{Q}$ is a symmetric positive definite matrix that all the eigenvalues are positive. Thus the product of all the eigenvalues $\det(\mathbf{Q})$ is also positive, ensuring an un-trivial solution $\mathbf{Q}$ for (28) [137]. The optimization problem is

$$\min_{\mathbf{Q}} \ln(\det(\mathbf{Q})) + \text{Trace}\{\mathbf{R} \mathbf{Q}\}, \quad (29)$$

where $\det(\cdot)$ is the determinant of a matrix, and $\ln(\cdot)$ is the natural logarithm. This optimization problem can be solved analytically using the derivative relative to $\mathbf{Q}$, for which the derivation is provided in [138]. The solution $\mathbf{Q}$ is the inverse of the covariance matrix or the precision matrix

$$\mathbf{Q} = \mathbf{R}^{-1}. \quad (30)$$

The well-known Graphical LASSO [139] is similar to the optimization problem in (29) with the sparsity imposed, which can be expressed as

$$\min_{\mathbf{Q}} \ln(\det \mathbf{Q}) + \text{Trace}\{\mathbf{R} \mathbf{Q}\} + \rho \|\mathbf{Q}\|_1. \quad (31)$$

B. Applications

Biological network graphs are diverse in the literature, and appropriate topology measures have been proposed for their physical significance or particular applications. In Section V-B1, we review biological studies that inferred graph topology from observed data using model-based methods that imposed assumptions on signal relations and underlying graph structures. In addition to the commonly used assumptions, we review studies

that accommodated assumptions adapted to specific objectives in Section V-B2. Section V-B3 reviews studies that recovered brain functional connectivity networks using graph learning and neural network techniques.

1) Graph Learning Inference: The model-based graph topology approach can reliably recover brain intrinsic networks, which can potentially boost the performance of the following tasks. Aggarwal *et al.* [140] built the functional connectivity of fMRI signals across ROIs of the brain for ASD patients using the graph model in (26) with the Frobenius norm of connection weights regularization added. In another fMRI functional connectivity study [141], the graph learning approach illustrated in (27) was used to reconstruct brain activities from the recorded BOLD signal and generate the Laplacian matrix of the functional network. This graph learning diagram was validated on the fMRI dataset of normal control and frontotemporal dementia. The recovered BOLD signal from the graph learning can reveal more details than the traditional Gaussian filtered signal. Compared with the network estimated using Pearson's correlation, the graph learning approach can reveal the modularity without thresholding, and it can discover more significant connections.

Wang *et al.* [142] applied the graph learning in (27) to construct an fMRI connectivity network, on which the graph Fourier transform was applied to extract low-pass, band-pass, and high-pass graph components to classify participants of different ages. The classification accuracy achieved outperformed the one that inferred network structure using geometric distance and correlation.

Tavildar *et al.* [143] inferred graph topology for the ECoG recordings from macaque monkeys. Each electrode was considered to be a network vertex, while the graph learning method was applied to coefficients of the auto-regressive representations of the signals. The assumption imposed on the graph modeling was the graph smoothness expressed in (27). The simulation data showed that the proposed graph learning method generated a connectivity network that was more similar to the ground truth graph structure than the coherence inference measure.

Shafipour *et al.* [144] proposed a method that learned topology from nonstationary graph signals, and it was modeled by local diffusion processes, which were governed by the polynomial graph-shift operator (GSO). The eigenvector of GSO can be inferred from graph diffusion filter estimation, and the eigenvalues of GSO were estimated by imposing sparsity constraints. The synthesized graph signal was generated by graph filtering of the brain network that models the density of anatomical connectivity. The proposed GSO graph inference method was applied to estimate brain graph eigenvectors of the simulated data, and the recovery error was computed against the number of observation samples. For the noisy signal, the signal can be well recovered with enough observations.

Sardellitti *et al.* [145] developed an approach that can learn a topology with graph basis contributing to the block sparsity, from which the Laplacian matrix was inferred that graph signals could be bandlimited. This graph topology inference method was tested on iEEG recording of one epilepsy patient. The pre-ictal and ictal graph network showed a dramatic difference that the network connections greatly decreased after the seizure offset.

2) Graph Learning With Objectives: For the graph topology inference methods, the most common graph structure assumptions were sparsity and smoothness. Additionally, studies have proposed graph learning methods with application-specific assumptions. Qiao *et al.* [146] learned the graph using the method given in (26) by incorporating the prior knowledge that the brain connectivity network tends to show modularity. The modularity can be modeled with the rank of the graph matrix since the strong modular graph structures tend to show sparsity with low ranks. The classification accuracy with functional connectivity network generated with the additional modularity constraint was 89.01% in discriminating MCI from normal controls, which outperformed the methods that adopted Pearson's correlation, the sparse only, and the low-rank only topology inference methods.

Graph learning procedures can be performed in a supervised or semi-supervised manner when labels or objectives are available. The commonly used unsupervised term is the consensus graph structure that can be learned across multiple graph networks.

Tan *et al.* [147] proposed a graph learning method to learn the consensus graph across multiple omics networks, lncRNA and disease similarity matrices, and available labels for the lncRNA and disease association. The optimization problem modeled can simultaneously generate the consensus graph and predict the lncRNA and disease association.

Turja *et al.* [148] proposed an approach to learn a consensus graph network from longitudinal connectivity networks by penalizing unreasonable neurological changes and preserving geometric structures along the time that fMRI data were recorded. Zhu *et al.* [149] developed a diagram to learn the consensus graph across multiple imaging modalities. Besides, they accommodated the available clinical labels and scores in the optimization procedure as the supervised term, which enhanced the following MCI subjects identification and MCI stages recognition tasks.

The consensus graph learning has been applied to control the disturbance of the monitoring system. Yang *et al.* [150] constructed graph networks for four elliptic parameters estimated from the chest and abdomen movements recorded by the Kinect depth camera at different time samples. The graph edges were the connections between different time samples measured using the Gaussian kernel function. The semi-supervised learning was performed by minimizing the variation of the network graph. For the signal contaminated by noise, the optimization of a fidelity term was included with the smoothness regularization term. They extended this graph learning approach denoising signals and smoothing graph networks to estimate heart rates from recorded depth video [151].

Graph learning approaches have been applied to improve the pooling step for GNNs. Padole *et al.* [152] learned the coarsening Laplacian matrix for GCN, which is analogous to the pooling step in CNNs that can preserve the topological structure of the original graph. This work performed the graph coarsening with two stages: first, the reduced graph Laplacian matrix was achieved by the graph wavelet transform; second, the coarsening Laplacian matrix was learned by constraining its eigenvectors to be an orthogonal matrix and minimizing the distance between

the coarsening Laplacian matrix and the eigenvector reduced Laplacian matrix.

Graph learning method has also been applied to estimate the source distribution for epileptic interictal spikes. Xuyen *et al.* [153] constructed the graph network with the covariance matrix of EEG channels, and added the smoothness constraint of the Laplacian matrix to the source localization method LORETA, which minimizes the distance between recorded data and the estimated source distribution. A simulation experiment in this study showed the proposed method can accurately localize epileptic sources with the proper sparsity.

3) Brain Functional Connectivity Learning: The brain functional connectivity network has been of interest in neuroscience, which models the brain activities as complex network interactions across different brain regions. The inference of the brain connectivity network may help us to understand interactions of brain regions regarding neurological disorders. GNNs can learn embeddings for brain functional connectivity networks, which can be further used to infer the underlying brain networks. In this section, we review studies that learned brain connectivity networks using GNNs. Various assumptions about the network structure or the representations residing on the node were imposed to infer brain networks for specific objectives.

Ni *et al.* [154] extracted features from fMRI data for each ROI to classify Choice Reaction Time tasks. The brain network inference was performed using the model-based GNN. The loss function was formulated with the maximum likelihood estimation of Markov Random Field modeled for connectivity edges. Instead of inferring brain networks from GNNs learned node representations, Ghoroghchian *et al.* [155] embedded the brain similarity matrix learning within the GNN procedure through the target probability distribution assumption based on the graph topology. The loss function was to minimize the Kullback-Leibler divergence between the inner product of the pairwise embeddings and presumed target probability distributions.

Instead of imposing assumptions on node embeddings, studies imposed assumptions on network structures to infer graph topologies. Zhang *et al.* [156] proposed a loop framework to learn embeddings from a random graph, on which a small-world structured network was generated and used as the graph structure. This loop was repeated until stable embeddings were generated. Lian *et al.* [157] constructed a graph network from EEG recordings with frequencies as graph nodes, and edge weights were calculated using the correlation coefficients. Based on this prior network, a GNN framework was proposed to jointly infer accurate graph topology and predict epileptic seizures by minimizing the loss function with available seizure labels.

VI. DISCUSSION

We have presented a systematic review of the literature about the application of GSP, GNN and graph learning in biology, for which graph network is a typical data structure. The collected methods in this review can potentially unveil intricacies of health and complex disease states in a computational way. This section summarizes the lessons learned, issues remained, and future directions for graph-based analysis methods.

A. Lessons Learned

The scopes and effectiveness of GSP, GNN, and graph inference approach applied to analyzing biological data summarized as follows:

- GSP sheds light on various biomedical engineering applications, such as BCI, heartbeat monitoring, for which feature extractions play a critical role in determining the following performance of classification, detection, and prediction. GSP can extract features in the graph domain and considers the problem as a network system, which has the potential to improve the performance of biomedical tasks.
- Compared to the isotropic Gaussian filter, the anatomically adapted graph filter can enhance the SNR of the fMRI signal. Furthermore, the vertex-frequency representations can project signals onto a localized vertex or a subnetwork at a specific graph frequency band, which could further improve the quality of fMRI data. They can potentially reveal the brain connectivity activities that used to be embedded in noises.
- GNNs are the extension of deep neural networks to graph-structured data. In biological research, graph networks are typical structures across different levels, from the population graph with patients as network nodes to the omics networks with biological molecules as nodes. GNNs have been shown to be a computational approach that can effectively predict disease-gene association, functions and structures of biological molecules, and drug-target interactions. Additionally, it can help the diagnosis of patients with neurological disorders through the population graph.
- Graph topology inference can reveal the underlying graph structure from data observed by imposing assumptions about the data relations or graph structures adapted to certain biological problems, such as omics networks and brain connectivity networks. Furthermore, the graph topology learning can be modeled together with available labels, which can jointly predict labels and the graph structure.
- GNNs can be used to infer graph topologies with biological objectives. In contrast, graph learning techniques can refine GNNs implementations, such as the pooling layer, which can coarsen the network without losing topological structures.

B. Challenges and Opportunities

The graph-based analysis methods, including GSP, GNN, and graph learning, are still under development. The technical challenges and opportunities, and their biological applications, are listed:

Invariant GFT.: The fundamental operation of GSP is GFT. GFT can be derived from the graph Laplacian matrix. However, the eigenvalues of the Laplacian matrix can have multiplicity, which contributes to multiple eigenvectors for one sub-frequency band of the graph domain. The multiplicity can lead to variant GFT and make the sub-frequency band decomposition irreproducible. The Jordan decomposition of the adjacency matrix leads to unique and invariant GFT [55], which however is

a numerically unstable process. An alternative technique establishes an explicit optimization procedure minimizing a cut size related quantity that can compute orthogonal graph basis [40]. The invariant GFT remains to be an active research topic in the community of GSP.

Inductive GNN.: Most studies that predicted node labels or associations for omics networks have been conducted using GNNs with transductive learning manner. The trained model cannot be generalized to unseen biological targets during training. To address this issue, an inductive learning GNN has been developed to generate a model by sampling and aggregating within the neighborhood that can learn embeddings for unseen data [68]. The inductive learning GNN methods can be applied to produce a general model trained on the available data and can be used for newly discovered biological targets without retraining procedure.

Directed graph. Additional problems arise to develop GSP operations for directed graphs, for which the Laplacian matrix is not symmetric, and thus the orthogonal eigendecomposition is not guaranteed. The non-symmetric Laplacian matrix makes the implementation of GNNs and graph topology inference approaches challenging as well. The selection of orthogonal graph basis according to an explicit optimization problem has been proposed to generate GFT for directed graphs [40]. This approach can be a future study direction to model the biological objectives together with GSP techniques to produce an adapted GFT basis that is optimal for specific applications and can potentially be applied to directed biological networks. Also, the topology inference for directed graph use of these techniques remains to be developed.

Dynamic graph structure.: Current GNNs and graph topology learning approaches mainly considered the underlying graph as a static structure, which is not realistic in many biological applications. For instance, brain connectivity networks change with participants' states or experimental conditions. To address this issue, the spatial-temporal GNN has been proposed to learn dynamic graph pattern [158]. Approaches adopting the states switch [159] and the time-window [160] have been proposed to learn the dynamic graph topologies. Furthermore, a multilayer network architecture was proposed to model dynamic graphs that the temporal evolution are modelled with interlayer nodes connections [161]. The multilayer network model can potentially enlighten the development of GNNs and graph inference methods for dynamic graphs.

High-dimensional multiway tensors. The development of GSP is currently limited to signal-dimensional graph networks without considering modulations with additional dimensions, such as time. It will be necessary to develop multiway GSP approaches that can deal with more complex real-life data that couple with multidimensional factors. Preliminary studies that extend single-modal GSP methods to the multiway domain have been proposed, which have been reviewed in [162].

VII. CONCLUSION

The development of GSP provides us with new sights into the graph-structured biological data. We conduct a systematic

literature review of biological data analysis in the graph domain, including GSP, GNN, and graph learning techniques. We further categorized them for specific biological applications. These approaches have been applied to different levels of biological data for diverse purposes. Even though there remain some issues in the graph methods, they are promising techniques for biological engineering and science. Especially, it can integrate multimodal data from clinical applications with biologically plausible interactions that can potentially unveil mechanisms of complex diseases.

VIII. SUPPLEMENTARY MATERIAL

A. Appendix 1 Study Selection Criteria

B. Appendix 2 Searching Fields for Each Database

IX. FIGURE LIST

Fig. 1: Methodology outline and application categorization.

Fig. 2: The Laplacian matrix eigenvalue decomposition of a graph network constructed from 3.36 s EEG recordings of a rodent brain. (a) The EEG electrodes placement and recordings of the rodent brain. (b) The graph network across five electrodes: the correlation coefficients. The five eigenvectors $\mathbf{u}_1$, $\mathbf{u}_2$, $\mathbf{u}_3$, $\mathbf{u}_4$, and $\mathbf{u}_5$ shown in (c) were generated from the Laplacian matrix of the network in (b), and they were ranked by the eigenvalues with the ascending order. For each eigenvector, its value distributions are demonstrated with the sensor index and the sensor locations, respectively.

Fig. 3: Signal-adapted graph wavelet frameworks. (a) Spline-based wavelet kernels. (b) Warped Translate wavelet kernels. (c) Histogram distribution of Laplacian eigenvalues of a graph signal. Vertical lines in (a) and (b) indicate the locations of eigenvalues.

Fig. 4: The deconvolution of fMRI with brain structural connectivity graph. The contour profiles of spatial filters with different adaptations to the heterogeneous orientation distribution of white matter fibers are shown in (a). (b) The isotropic Gaussian filter without adaptations to anatomical morphology nor the underlying white matter structure. (c) A filter adapted to the anatomical morphology, but not the white matter structural orientations. (d) A graph filter adapted to both the anatomical morphology and the underlying structure (cite Fig. 1 in [41], copyright has been approved by IEEE).

Fig. 5: Different levels of GNN learning: node, edge, and graph. GNNs can be applied to perform node-wise, edge-wise (interaction prediction), and graph-wise learning for different biological data applications.

Fig. 6: Inductive learning and transductive learning for node-wise classification. Inductive learning tries to learn a general model from available instances, which can perform inference for the unseen instances. In contrast, transductive learning is trained on both labeled and unlabelled samples, and the learned model can assign labels for unlabelled instances.

Fig. 7: The application of GNNs on population graphs with multimodal data. (a) The graph was constructed with imaging data and metadata as node features and edge weights. (b) The multiple graphs were generated by metadata and their correlated

features. GCN was applied onto each graph, and the learned embeddings were concatenated for the following prediction procedure using long short-term memory [89]. (c) The graph ensemble was generated by the random edge drops from the multimodal population graph. The prediction tasks were performed by the consensus or the average across all the GCNs in the ensemble [90].

Fig. 8: Interaction prediction framework. This framework can propagate edge labels during transductive learning.

Fig. 9: Interaction prediction framework for DTIs. In this framework, representations for proteins and chemical compounds are learned separately, and their representations can be concatenated for their association prediction.

Fig. 10: GNNs across multiple graphs. (a) Independent GNN. GNN was applied to multiple graph networks independently to generate graph embeddings that were concatenated for the following tasks. (b) Interactive GNN. GNN applied to multiple graphs interactively with either their learning weights or layer outputs shared. (c) Hierarchical GNN. Multiple graphs were modeled as nodes of a hierarchical graph, for which GNN was applied onto the node graph and hierarchical graph to learn embeddings for a specific task.

REFERENCES

- [1] Y. Hasin, "Multi-omics approaches to disease," *Genome Biol.*, vol. 18, no. 83, p. 1, May 2017.
- [2] N. Naidoo, Y. Pawitan, R. Soong, D. N. Cooper, and C.-S. Ku, "Human genetics and genomics a decade after the release of the draft sequence of the human genome," *Hum. Genomic.*, vol. 5, no. 6, p. 577, Oct. 2011.
- [3] C. E. Cook, M. T. Bergman, R. D. Finn, G. Cochran, E. Birney, and R. Apweiler, "The european bioinformatics institute in 2016: Data growth and integration," *Nucleic Acids Res.*, vol. 44, no. D1, pp. D 20–D26, Dec. 2016.
- [4] N. Ganapathy, R. Swaminathan, and T. M. Deserno, "Deep learning on 1-D biosignals: A taxonomy-based survey," *Yearbook Med. Informat.*, vol. 27, no. 1, pp. 98–109, Aug. 2018.
- [5] J. Schl pfer and H. J. Wellens, "Computer-interpreted electrocardiograms: Benefits and limitations," *J. Amer. College Cardiol.*, vol. 70, no. 9, pp. 1183–1192, Aug. 2017.
- [6] B. Rim, N.-J. Sung, S. Min, and M. Hong, "Deep learning in physiological signal data: A survey," *Sensors*, vol. 20, no. 4, p. 969, Feb. 2020.
- [7] V. Marx, "Biology: The big challenges of Big Data," *Nature*, vol. 498, no. 7453, pp. 255–260, Jun. 2013.
- [8] S. Boccaletti, V. Latora, Y. Moreno, M. Chavez, and D.-U. Hwang, "Complex networks: Structure and dynamics," *Phys. Rep.*, vol. 424, no. 4–5, pp. 175–308, Feb. 2006.
- [9] L. d. F. Costa, F. A. Rodrigues, G. Traverso, and P. R. V. Boas, "Characterization of complex networks: A survey of measurements," *Adv. Phys.*, vol. 56, no. 1, pp. 175–308, Apr. 2007.
- [10] B. Bollob s, *Modern Graph Theory*, F. G. S. Axler and K. Ribet, Eds. New York: Springer, 2013.
- [11] M. Rubinov and O. Sporns, "Complex network measures of brain connectivity: Uses and interpretations," *Neuroimage*, vol. 52, no. 3, pp. 1059–1069, Sep. 2010.
- [12] T. Ideker, T. Galitski, and L. Hood, "A new approach to decoding life: Systems biology," *Annu. Rev. Genomic. Hum. Genet.*, vol. 2, no. 1, pp. 343–372, Sep. 2001.
- [13] A. Ortega, P. Frossard, J. Kova evi , J. M. Moura, and P. Vandergheynst, "Graph signal processing: Overview, challenges, and applications," *Proc. IEEE Proc. IRE*, vol. 106, no. 5, pp. 808–828, May 2018.
- [14] A. Sandryhaila and J. M. Moura, "Discrete signal processing on graphs," *IEEE Trans. Signal Process.*, vol. 61, no. 7, pp. 1644–1656, Apr. 2013.
- [15] A. Sandryhaila and J. M. F. Moura, "Discrete signal processing on graphs: Frequency analysis," *IEEE Trans. Signal Process.*, vol. 62, no. 12, pp. 3042–3054, Jun. 2014.
- [16] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.

- [17] H. Behjat and M. Larsson, "Spectral characterization of functional MRI data on voxel-resolution cortical graphs," in *Proc. IEEE 17th Int. Symp. Biomed. Imag.*, Iowa City, IA, USA, 2020, pp. 558–562.
- [18] T. A. W. Bolton, Y. Farouj, M. Inan, and D. V. D. Ville, "Structurally-informed deconvolution of functional magnetic resonance imaging data," in *Proc. IEEE 16th Int. Symp. Biomed. Imag.*, Venice, Italy, 2019, pp. 1545–1549.
- [19] A. Brahim, M. H. E. Hassani, and N. Farrugia, "Classification of autism spectrum disorder through the graph fourier transform of fMRI temporal signals projected on structural connectome," in *Proc. Int. Conf. Comput. Anal. Images Patterns*, 2019, pp. 45–55.
- [20] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, May 2013.
- [21] A. G. Marques, N. Kiyavash, J. M. Moura, D. V. D. Ville, and R. Willett, "Graph signal processing: Foundations and emerging directions," *IEEE Signal Process. Mag.*, vol. 37, no. 6, pp. 11–13, Nov. 2020.
- [22] F. Gama, E. Isufi, G. Leus, and A. Ribeiro, "Graphs, convolutions, and neural networks: From graph filters to graph neural networks," *IEEE Signal Process. Mag.*, vol. 37, no. 6, pp. 128–138, Nov. 2020.
- [23] J. Zhou et al., "Graph neural networks: A review of methods and applications," 2018, *arXiv:1812.08434*.
- [24] X. Dong, D. Thanou, M. Rabbat, and P. Frossard, "Learning graphs from data: A signal representation perspective," *IEEE Signal Process. Mag.*, vol. 36, no. 3, pp. 44–63, May 2019.
- [25] W. Zhang, J. Chien, J. Yong, and R. Kuang, "Network-based machine learning and graph theory algorithms for precision oncology," *NPJ Precis. Oncol.*, vol. 1, no. 1, pp. 1–15, Aug. 2017.
- [26] I. L. Hudson, "Data integration using advances in machine learning in drug discovery and molecular biology," in *Artificial Neural Networks*, H. Cartwright, Ed. New York, NY, US: Springer, 2021, pp. 167–184.
- [27] S. Mallik and Z. Zhao, "Graph- and rule-based learning algorithms: A comprehensive review of their applications for cancer type classification and prognosis using genomic data," *Brief. Bioinf.*, vol. 21, no. 2, pp. 368–394, Mar. 2019.
- [28] M. Sun, S. Zhao, C. Gilvary, O. Elemento, J. Zhou, and F. Wang, "Graph convolutional networks for computational drug development and discovery," *Brief. Bioinf.*, vol. 21, no. 3, pp. 919–935, May 2020.
- [29] W. Huang, T. A. Bolton, J. D. Medaglia, D. S. Bassett, A. Ribeiro, and D. V. D. Ville, "A graph signal processing perspective on functional brain imaging," *Proc. IEEE Proc. IRE*, vol. 106, no. 5, pp. 868–885, May 2018.
- [30] L. S. Qiao, L. M. Zhang, S. C. Chen, and D. G. Shen, "Data-driven graph construction and graph learning: A review," *Neurocomputing*, vol. 312, pp. 336–351, Oct. 2018.
- [31] S. Jin, X. Zeng, F. Xia, W. Huang, and X. Liu, "Application of deep learning methods in biological networks," *Brief. Bioinf.*, vol. 22, no. 2, pp. 1902–17, May 2020.
- [32] D. Cvetković, "Spectral recognition of graphs," *Yugoslav J. Operations Res.*, vol. 22, no. 2, pp. 145–161, 2012.
- [33] P. M. Casillas-Espinosa et al., "Harmonization of the pipeline for seizure detection to phenotype post-traumatic epilepsy in a preclinical multicenter study on post-traumatic epileptogenesis," *Epilepsy Res.*, vol. 156, pp. 106–131, Oct. 2019.
- [34] D. I. Shuman, B. Ricaud, and P. Vandergheynst, "Vertex-frequency analysis on graphs," *Appl. Comput. Harmon. Anal.*, vol. 40, no. 2, pp. 260–291, Mar. 2016.
- [35] D. C. V. Essen et al., "The WU-minn human connectome project: An overview," *Neuroimage*, vol. 80, pp. 62–79, Oct. 2013.
- [36] H. Behjat and D. V. D. Ville, "Spectral design of signal-adapted tight frames on graphs," in *Vertex-Frequency Analysis of Graph Signals*, L. Stanković and E. Sejdić, Eds. Cham: Springer, 2019, pp. 177–206.
- [37] D. I. Shuman, C. Wiesmeyer, N. Holighaus, and P. Vandergheynst, "Spectrum-adapted tight graph wavelet and vertex-frequency frames," *IEEE Trans. Signal Process.*, vol. 63, no. 16, pp. 4223–4235, Aug. 2015.
- [38] D. V. D. Ville, R. Demesmaeker, and M. G. Preti, "When slepian meets fiedler: Putting a focus on the graph spectrum," *IEEE Signal Process. Lett.*, vol. 24, no. 7, pp. 1001–1004, Jul. 2017.
- [39] M. Petrovic, T. A. Bolton, M. G. Preti, R. Liégeois, and D. V. D. Ville, "Guided graph spectral embedding: Application to the c. elegans connectome," *Netw. Neurosci.*, vol. 3, no. 3, pp. 807–826, Nov. 2019.
- [40] S. Sardellitti, S. Barbarossa, and P. D. Lorenzo, "On the graph fourier transform for directed graphs," *IEEE J. Sel. Topics Signal Process.*, vol. 11, no. 6, pp. 796–811, Jul. 2017.
- [41] D. Abramian, M. Larsson, A. Eklund, and H. Behjat, "Improved functional mri activation mapping in white matter through diffusion-adapted spatial filtering," in *Proc. IEEE 17th Int. Symp. Biomed. Imag.*, Iowa City, IA, USA, 2020, pp. 539–543.
- [42] J. D. Medaglia et al., "Functional alignment with anatomical networks is associated with cognitive flexibility," *Nature Hum. Behav.*, vol. 2, no. 2, pp. 156–164, Feb. 2018.
- [43] S. Sihag, S. Naze, F. Taghdiri, M. C. Tartaglia, and J. R. Kozloski, "GSP analysis of brain imaging data from athletes with history of multiple concussions," in *Proc. IEEE Glob. Conf. Signal Inf. Process.*, Ottawa, ON, Canada, 2019, pp. 1–5.
- [44] S. Sihag et al., "Multimodal dynamic brain connectivity analysis based on graph signal processing for former athletes with history of multiple concussions," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 6, pp. 284–299, 2020.
- [45] Y. Li and G. Mateos, "Identifying structural brain networks from functional connectivity: A network deconvolution approach," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Brighton, United Kingdom, 2019, pp. 1135–1139.
- [46] M. G. Preti and D. V. D. Ville, "Graph slepians to probe into large-scale network organization of resting-state functional connectivity," in *Proc. 51st Asilomar Conf. Signals, Syst., Comput.*, Pacific Grove, CA, USA, 2017, pp. 1539–1543.
- [47] T. A. W. Bolton, Y. Farouj, S. Obertino, and D. V. D. Ville, "Graph slepians to strike a balance between local and global network interactions: Application to functional brain imaging," in *Proc. IEEE 15th Int. Symp. Biomed. Imag.*, Washington, DC, USA, 2018, pp. 1239–1243.
- [48] K. Georgiadis, N. Laskaris, S. Nikolopoulos, and I. Kompatsiaris, "Connectivity steered graph fourier transform for motor imagery BCI decoding," *J. Neural Eng.*, vol. 16, no. 5, Aug. 2019.
- [49] N. Ghoroghchian, S. C. Draper, and R. Genov, "A hierarchical graph signal processing approach to inference from spatiotemporal signals," in *Proc. 29th Biennial Symp. Commun.*, Toronto, ON, Canada, 2018, pp. 1–5.
- [50] G. Kalantar, H. Sadreazami, A. Mohammadi, and A. Asif, "Adaptive dimensionality reduction method using graph-based spectral decomposition for motor imagery-based brain-computer interfaces," in *Proc. IEEE Glob. Conf. Signal Inf. Process.*, Montreal, QC, Canada, 2017, pp. 990–994.
- [51] T. Maloney, G. Kalantar, and A. Mohammadi, "Progressive fusion of multi-rate motor imagery classification for brain computer interfaces," in *Proc. IEEE 60th Int. Midwest Symp. Circuits Syst.*, Boston, MA, USA, 2017, pp. 1613–1616.
- [52] P. Mathur and V. K. Chakka, "Graph signal processing of eeg signals for detection of epilepsy," in *Proc. 7th Int. Conf. Signal Process. Integr. Netw.*, Noida, India, 2020, pp. 839–843.
- [53] L. Rui, H. Nejati, and N. M. Cheung, "Dimensionality reduction of brain imaging data using graph signal processing," in *Proc. IEEE Int. Conf. Image Process.*, Phoenix, AZ, USA, 2016, pp. 1329–1333.
- [54] J. Wang, V. D. Calhoun, J. M. Stephen, T. W. Wilson, and Y. P. Wang, "Integration of network topological features and graph Fourier transform for fMRI data analysis," in *Proc. IEEE 15th Int. Symp. Biomed. Imag.*, Washington, DC, USA, Apr. 2018, pp. 92–96.
- [55] H. Padole, S. D. Joshi, and T. K. Gandhi, "Early detection of alzheimer's disease using graph signal processing on neuroimaging data," in *Proc. 2nd Eur. Conf. Elect. Eng. Comput. Sci.*, Bern, Switzerland, 2018, pp. 302–306.
- [56] S. Mortaheb et al., "A graph signal processing approach to study high density EEG signals in patients with disorders of consciousness," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2019, pp. 4549–4553.
- [57] A. D. Choudhury and A. S. Chowdhury, "Change: Cardiac health analysis using graph eigenvalues," in *Proc. 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Honolulu, HI, USA, 2018, pp. 4025–4029.
- [58] S. Itani and D. Thanou, "Combining anatomical and functional networks for neuropathology identification: A case study on autism spectrum disorder," *Med. Image Anal.*, vol. 69, Apr. 2021, Art. no. 101986.
- [59] Y. Y. Pilavci and N. Farrugia, "Spectral graph wavelet transform as feature extractor for machine learning in neuroimaging," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Brighton, U. K., 2019, pp. 1140–1144.
- [60] L. Goldsberry, W. Huang, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Brain signal analytics from graph signal processing perspective," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, New Orleans, LA, USA, 2017, pp. 851–855.

- [61] W. Huang, L. Goldsberry, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Graph frequency analysis of brain signals," *IEEE J. Sel. Top. Signal Process.*, vol. 10, no. 7, pp. 1189–1203, Oct. 2016.
- [62] J. J. Shih, D. J. Krusienski, and J. R. Wolpaw, "Brain-computer interfaces in medicine," *Mayo Clin. Proc.*, vol. 87, no. 3, pp. 268–279, 2012.
- [63] L. Lacasa, F. B. B. Luque, J. Luque, and J. C. Nuno, "From time series to complex networks: The visibility graph," *Proc. Nat. Acad. Sci.*, vol. 105, no. 13, pp. 4972–4975, Apr. 2008.
- [64] G. Angius, D. Barcellona, E. Cauli, L. Meloni, and L. Raffo, "Myocardial infarction and antiphospholipid syndrome: A first study on finger PPG waveforms effects," in *Comput. Cardiol.*, Sep. 2012, pp. 517–520.
- [65] R. Banerjee, R. Vempada, K. Mandana, A. D. Choudhury, and A. Pal, "Identifying coronary artery disease from photoplethysmogram," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.: Adjunct.*, 2016, pp. 1084–1088.
- [66] D. D. Nguyen and G. W. Wei, "AGL-Score: Algebraic graph learning score for protein-ligand binding scoring, ranking, docking, and screening," *J. Chem. Inf. Model.*, vol. 59, no. 7, pp. 3291–3304, Jun. 2019.
- [67] M. G. Preti and D. V. D. Ville, "Decoupling of brain function from structure reveals regional behavioral specialization in humans," *Nature Commun.*, vol. 10, no. 1, pp. 1–7, Oct. 2017.
- [68] I. Jestrović, J. L. Coyle, and E. Sejdić, "Differences in brain networks during consecutive swallows detected using an optimized vertex-frequency algorithm," *Neuroscience*, vol. 344, pp. 113–123, Mar. 2017.
- [69] M. Sun, E. Isufi, N. M. S. de Groot, and R. C. Hendriks, "Graph-time spectral analysis for atrial fibrillation," *Biomed. Signal Process. Control*, vol. 59, May 2020, Art. no. 101915.
- [70] J. Bruna, W. Zaremba, A. Szlam, and Y. LeCun, "Spectral networks and locally connected networks on graphs," 2013, *arXiv:1312.6203*.
- [71] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Adv. Neural Inf. Process. Syst.*, Dec. 2016, pp. 3844–3852.
- [72] D. K. Hammon, P. Vandergheynst, and R. Gribonval, "Wavelets on graphs via spectral graph theory," *Appl. Comput. Harmon. Anal.*, vol. 30, no. 2, pp. 129–150, Mar. 2011.
- [73] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
- [74] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 1025–1035.
- [75] P. Veličković, A. Casanova, P. Liò, G. Cucurull, A. Romero, and Y. Bengio, "Graph attention networks," in *Proc. Int. Conf. Learn. Representations*, 2018. [Online]. Available: <https://openreview.net/forum?id=rJXMpikCZ>
- [76] E. Isufi, F. Gama, and A. Ribeiro, "EdgeNets: Edge varying graph neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2021, doi: [10.1109/TPAMI.2021.3111054](https://doi.org/10.1109/TPAMI.2021.3111054).
- [77] J. Atwood and D. Towsley, "Diffusion-convolutional neural networks," in *Adv. Neural Inf. Process. Syst.*, Dec. 2016, pp. 1993–2001.
- [78] D. Zhang and M. R. Kabuka, "Protein family classification with multi-layer graph convolutional networks," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, Madrid, Spain, 2018, pp. 2390–2393.
- [79] H. Xue, X. K. Sun, and W. X. Sun, "Multi-hop hierarchical graph neural networks," in *Proc. IEEE Int. Conf. Big Data Smart Comput.*, Busan, Korea (South), 2020, pp. 82–89.
- [80] N. Ravindra, A. Sehanobish, J. L. Pappalardo, D. A. Hafler, and D. V. Dijk, "Disease state prediction from single-cell data using graph attention networks," in *Proc. ACM Conf. Health, Inference, Learn.*, 2020, pp. 121–130.
- [81] R. Schulte-Sasse, S. Budach, D. Hnisch, and A. Marsico, "Graph convolutional networks improve the prediction of cancer driver genes," in *Proc. Int. Conf. Artif. Neural Netw.*, 2019, pp. 658–668.
- [82] W. Zhao, J. Zhu, M. Yang, D. Xiao, G. P. C. Fung, and X. Chen, "A semi-supervised network embedding model for protein complexes detection," in *Proc. 32nd AAAI Conf. Artif. Intell.*, 2018, pp. 8185–8186.
- [83] Y. Zhang, Y. Chen, and T. Hu, "PANDA: Prioritization of autism-genes using network-based deep-learning approach," *Genet. Epidemiol.*, vol. 44, no. 4, pp. 382–394, Feb. 2020.
- [84] V. N. Ioannidis, A. G. Marques, and G. B. Giannakis, "Graph neural networks for predicting protein functions," in *Proc. IEEE 8th Int. Workshop Comput. Adv. Multi-Sensor Adaptive Process.*, Le Gosier, Guadeloupe, Guadeloupe, 2019, pp. 221–225.
- [85] X. Pan and H. B. Shen, "Inferring disease-associated microRNAs using semi-supervised multi-label graph convolutional networks," *iScience*, vol. 20, pp. 265–277, Oct. 2019.
- [86] J. Liu, D. Zeng, M. Lu, and J. Wang, "Mild cognitive impairment identification based on multi-view graph convolutional networks," in *Proc. 7th Int. Conf. Adv. Cloud Big Data*, Suzhou, China, China, 2019, pp. 309–314.
- [87] X. G. Song, A. Elazab, and Y. X. Zhang, "Classification of mild cognitive impairment based on a combined high-order network and graph convolutional network," *IEEE Access*, vol. 8, pp. 42 816–42827, Feb. 2020.
- [88] S. Yu, G. Yue, A. Elazab, X. Song, T. Wang, and B. Lei, "Multi-scale graph convolutional network for mild cognitive impairment detection," in *Proc. Int. Workshop Graph Learn. Med. Imag.*, 2019, pp. 79–87.
- [89] X. Zhao, F. Zhou, L. Ou-Yang, T. Wang, and B. Lei, "Graph convolutional network analysis for mild cognitive impairment prediction," in *Proc. IEEE 16th Int. Symp. Biomed. Imag.*, Venice, Italy, 2019, pp. 1598–1601.
- [90] J. Valençon and M. Coates, "Multiple-graph recurrent graph convolutional neural network architectures for predicting disease outcomes," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Brighton, U. K., 2019, pp. 3157–3161.
- [91] R. Anirudh and J. J. Thiagarajan, "Bootstrapping graph convolutional neural networks for autism spectrum disorder classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Brighton, United Kingdom, 2019, pp. 3197–3201.
- [92] C. Wang *et al.*, "A cancer survival prediction method based on graph convolutional network," *IEEE Trans. Nanobiosci.*, vol. 19, no. 1, pp. 117–126, Jan. 2020.
- [93] A. Fout, J. Byrd, B. Shariats, and A. Ben-Hur, "Protein interface prediction using graph convolutional networks," in *Adv. Neural Inf. Process. Syst.*, Dec. 2017, pp. 6531–6540.
- [94] L. Liu *et al.*, "Integrating sequence and network information to enhance protein-protein interaction prediction using graph convolutional networks," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, San Diego, CA, USA, 2019, pp. 1762–1768.
- [95] T. Zhao, Y. Hu, J. Peng, and L. Cheng, "GCN-CNN: A novel deep learning method for prioritizing lncRNA target genes," *Bioinformatics*, vol. 28, p. 28, Aug. 2020.
- [96] P. Xuan, S. Pan, T. Zhang, Y. Liu, and H. Sun, "Graph convolutional network and convolutional neural network based method for predicting lncRNA-disease associations," *Cells*, vol. 8, no. 9, Aug. 2019, Art. no. 1012.
- [97] R. Cai, X. Chen, Y. Fang, M. Wu, and Y. Hao, "Dual-dropout graph convolutional network for predicting synthetic lethality in human cancers," *Bioinformatics*, vol. 36, no. 16, pp. 4458–4465, Aug. 2020.
- [98] P. Han *et al.*, "GCN-MF: Disease-gene association identification by graph convolutional networks and matrix factorization," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, Anchorage, AK, USA, 2019, pp. 705–713.
- [99] C. Li, H. Liu, Q. Hu, J. Que, and J. Yao, "A novel computational model for predicting microRNA-disease associations based on heterogeneous graph convolutional networks," *Cells*, vol. 8, no. 9, p. 977, Aug. 2019.
- [100] L. Wang, Z. H. You, Y. M. Li, K. Zheng, and Y. A. Huang, "GCNCDA: A new method for predicting circRNA-disease associations based on graph convolutional network algorithm," *PLoS Comput. Biol.*, vol. 16, no. 5, May 2020, Art. no. e 1007568.
- [101] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, vol. 34, no. 13, pp. i 457–i466, Jul. 2018.
- [102] H. E. Manoochehri, A. Pillai, and M. Nourani, "Graph convolutional networks for predicting drug-protein interactions," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, San Diego, CA, USA, 2019, pp. 1223–1225.
- [103] M. Tsubaki, K. Tomii, and J. Sese, "Compound-protein interaction prediction with end-to-end learning of neural networks for graphs and sequences," *Bioinformatics*, vol. 35, no. 2, pp. 309–318, Jan. 2019.
- [104] X. Wang, Y. Liu, F. Lu, H. Li, P. Gao, and D. Wei, "Dipeptide frequency of word frequency and graph convolutional networks for DTA prediction," *Front. Bioeng. Biotechnol.*, vol. 8, p. 267, Apr. 2020.
- [105] Z. Quan, Y. Guo, X. Lin, Z. J. Wang, and X. Zeng, "GraphCPI: Graph neural representation learning for compound-protein interaction," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, San Diego, CA, USA, 2019, pp. 717–722.
- [106] T. Tanebe and T. Ishida, "End-to-end learning based compound activity prediction using binding pocket information," in *Proc. Int. Conf. Intell. Comput.*, 2019, pp. 226–234.
- [107] G. Zhou, J. Wang, X. Zhang, and G. Yu, "Deepgo: Predicting gene ontology annotations of proteins via graph convolutional network," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, San Diego, CA, USA, 2019, pp. 1836–1841.

- [108] J. Lim, S. Ryu, K. Park, Y. J. Choe, J. Ham, and W. Y. Kim, "Predicting drug-target interaction using a novel graph neural network with 3D structure-embedded graph representation," *J. Chem. Inf. Model.*, vol. 59, no. 9, pp. 3981–3988, Aug. 2019.
- [109] E. Wang et al., "A graph convolutional network-based method for chemical-protein interaction extraction: Algorithm development," *JMIR Med. Inform.*, vol. 8, no. 5, May 2020, Art. no. e17643.
- [110] A. Atamna, N. Sokolovska, and J. C. Crivello, "A principled approach to analyze expressiveness and accuracy of graph neural networks," in *Proc. Int. Symp. Intell. Data Anal.*, 2020, pp. 27–39.
- [111] C. Y. Wee, C. Liu, A. Lee, J. S. Poh, H. Ji, and A. Qiu, "Cortical graph neural network for ad and MCI diagnosis and transfer learning across populations," *NeuroImage: Clin.*, vol. 23, Jul. 2019, Art. no. 101929.
- [112] S. Lee, S. Lim, T. Lee, I. Sung, and S. Kim, "Cancer subtype classification and modeling by pathway attention and propagation," *Bioinformatics*, vol. 36, no. 12, pp. 3918–3824, Mar. 2020.
- [113] Y. Xie, M. Gong, Y. Gao, A. K. Qin, and X. Fan, "A multi-task representation learning architecture for enhanced graph classification," *Front. Neurosci.*, vol. 13, Jan. 2019, Art. no. 1395.
- [114] X. Li, N. C. Dvornek, Y. Zhou, J. Zhuang, P. Ventola, and J. S. Duncan, "Graph neural network for interpreting task-fMRI biomarkers," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2019, pp. 485–493.
- [115] Y. Zhang, L. Zhan, W. Cai, P. Thompson, and H. Huang, "Integrating heterogeneous brain networks for predicting brain disease conditions," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2019, pp. 214–222.
- [116] Y. Guo, H. Nejati, and N. M. Cheung, "Deep neural networks on graph signals for brain imaging analysis," in *Proc. IEEE Int. Conf. Image Process.*, Beijing, China, 2018, pp. 3295–3299.
- [117] X. H. Wang, T. Zhang, X. M. Xu, L. Chen, X. F. Xing, and C. L. P. Chen, "EEG emotion recognition using dynamical graph convolutional neural networks and broad learning system," in *Proc. IEEE Int. Conf. Bioinf. Biomed.*, 2018, pp. 1240–1244.
- [118] D. Yao et al., "Triplet graph convolutional network for multi-scale analysis of functional connectivity using functional MRI," in *Proc. Int. Workshop Graph Learn. Med. Imag.*, 2019, pp. 70–78.
- [119] J. Li, H. Meng, Y. Rong, W. Huang, H. Cheng, and J. Huang, "Semi-supervised graph classification: A hierarchical graph perspective," in *Proc. World Wide Web Conf.*, 2019, pp. 972–982.
- [120] G. Ma et al., "Deep graph similarity learning for brain data analysis," in *Proc. Int. Conf. Inf. Knowl. Manage.*, Beijing, China, 2019, pp. 2743–2751.
- [121] K. M. Borgwardt, C. S. Ong, S. Schöner, S. Vishwanathan, A. J. Smola, and H.-P. Kriegel, "Protein function prediction via graph kernels," *Bioinformatics*, vol. 21, no. "suppl_1", pp. i 47–i56, Jun. 2005.
- [122] P. Pati et al., "Hact-net: A hierarchical cell-to-tissue graph neural network for histopathological image classification," in *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging and Graphs in Biomedical Image Analysis*. Cham, Switzerland: Springer, 2020, pp. 208–219.
- [123] D. Jiménez-Sánchez, M. Ariz, and C. Ortiz-De-Solorzano, "Unsupervised learning of contextual information in multiplex immunofluorescence tissue cytometry," in *Proc. IEEE 17th Int. Symp. Biomed. Imag.*, Iowa City, IA, USA, 2020, pp. 1275–1279.
- [124] H. Yang, X. Zhen, Y. Chi, L. Zhang, and X.-S. Hua, "Cpr-gcn: Conditional partial-residual graph convolutional network in automated anatomical labeling of coronary arteries," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 3803–3811.
- [125] X. Yu, S.-H. Wang, and Y.-D. Zhang, "CGnet: A graph-knowledge embedded convolutional neural network for detection of pneumonia," *Inf. Process. Manage.*, vol. 58, no. 1, 2021, Art. no. 102411.
- [126] S. Y. Shin, S. Lee, I. D. Yun, and K. M. Lee, "Deep vessel segmentation by learning graphical connectivity," *Med. Image Anal.*, vol. 58, 2019, Art. no. 101556.
- [127] Z. Han, B. Wei, X. Xi, B. Chen, Y. Yin, and S. Li, "Unifying neural learning and symbolic reasoning for spinal medical report generation," *Med. Image Anal.*, vol. 67, 2021, Art. no. 101872.
- [128] L. Hansen and M. P. Heinrich, "Graphregnet: Deep graph regularisation networks on sparse keypoints for dense registration of 3D lung CTs," *IEEE Trans. Med. Imag.*, vol. 40, no. 9, pp. 2246–2257, Sep. 2021.
- [129] Z. Tian et al., "Interactive prostate mr image segmentation based on convlstm and GGNN," *Neurocomputing*, vol. 438, pp. 84–93, 2021.
- [130] V. Cantoni, A. Ferone, A. Petrosino, and G. S. di Baja, "A supervised approach to 3D structural classification of proteins," in *Proc. Int. Conf. Image Anal. Process.*, A. Petrosino, L. Maddalena, and P. Pala, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, pp. 326–335.
- [131] C. Donnat and S. Holmes, "Tracking network dynamics: A survey using graph distances," *Ann. Appl. Statist.*, vol. 12, no. 2, pp. 971–1012, Jun. 2018.
- [132] M. Menoret, N. Farrugia, B. Pasdeloup, and V. Gripon, "Evaluating graph signal processing for neuroimaging through classification and dimensionality reduction," in *Proc. IEEE Glob. Conf. Signal Inf. Process.*, Montreal, QC, Canada, 2018, pp. 618–622.
- [133] N. Meinshausen and P. Bühlmann, "High-dimensional graphs and variable selection with the lasso," *Ann. Statist.*, vol. 34, no. 3, pp. 1436–1462, 2006.
- [134] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Roy. Stat. Soc.: Ser. B. (Methodological)*, vol. 58, no. 1, pp. 267–288, 1996.
- [135] X. Dong, D. Thanou, P. Frossard, and P. Vandergheynst, "Learning laplacian matrix in smooth graph signal representations," *IEEE Trans. Signal Process.*, vol. 64, no. 23, pp. 6160–6173, Dec. 2016.
- [136] C. Hu, L. Cheng, J. Sepulcre, G. El Fakhri, Y. M. Lu, and Q. Li, "A graph theoretical regression model for brain connectivity learning of alzheimer's disease," in *Proc. IEEE 10th Int. Symp. Biomed. Imag.*, 2013, pp. 616–619.
- [137] M. Wainwright and M. Jordan, "Log-determinant relaxation for approximate inference in discrete markov random fields," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2099–2109, Jun. 2006.
- [138] L. Stankovic et al., "Graph signal processing-part III: Machine learning on graphs, from graph topology to applications," 2020, *arXiv:2001.00426*.
- [139] J. Friedman, T. Hastie, and R. Tibshirani, "Sparse inverse covariance estimation with the graphical lasso," *Biostatistics*, vol. 9, no. 3, pp. 432–441, 2008.
- [140] P. Aggarwal and A. Gupta, "Multivariate graph learning for detecting aberrant connectivity of dynamic brain networks in autism," *Med. Image Anal.*, vol. 56, pp. 11–25, 2019.
- [141] M. Kim, A. Moussa, P. Liang, D. Kaufer, P. J. Larenti, and G. Wu, "Revealing functional connectivity by learning graph laplacian," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2019, pp. 718–726.
- [142] J. Wang, L. Xiao, T. W. Wilson, J. M. Stephen, V. D. Calhoun, and Y. P. Wang, "Examining brain maturation during adolescence using graph laplacian learning based fourier transform," *J. Neurosci. Methods*, vol. 338, May 2020, Art. no. 108649.
- [143] S. Tavildar et al., "Inferring cortical connectivity from ecog signals using graph signal processing," *IEEE Access*, vol. 7, pp. 109 349–109362, Aug. 2019.
- [144] R. Shafipour, S. Segarra, A. G. Marques, and G. Mateos, "Network topology inference from non-stationary graph signals," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, New Orleans, LA, USA, 2017, pp. 5870–5874.
- [145] S. Sardellitti, S. Barbarossa, and P. D. Lorenzo, "Graph topology inference based on sparsifying transform learning," *IEEE Trans. Signal Process.*, vol. 67, no. 7, pp. 1712–1727, Apr. 2019.
- [146] L. Qiao, H. Zhang, M. Kim, S. T. L. Zhang, and D. Shen, "Estimating functional brain networks by incorporating a modularity prior," *NeuroImage*, vol. 141, pp. 399–407, Nov. 2016.
- [147] H. Tan et al., "Multiview consensus graph learning for lncRNA-disease association prediction," *Front. Genet.*, vol. 8, Feb. 2020, Art. no. 89.
- [148] M. A. Turja, L. C. P. Zsembik, G. Wu, and M. Styner, "Constructing consistent longitudinal brain networks by group-wise graph learning," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Interv.*, 2019, pp. 654–662.
- [149] Y. Zhu, X. Zhu, M. Kim, D. Kaufer, and G. Wu, "A novel dynamic hypergraph inference framework for computer assisted diagnosis of neuro-diseases," in *Proc. Int. Conf. Inf. Process. Med. Imag.*, 2017, pp. 158–169.
- [150] C. Yang, Y. Mao, G. Cheung, V. Stankovic, and K. Chan, "Graph-based depth video denoising and event detection for sleep monitoring," in *Proc. IEEE 16th Int. Workshop Multimedia Signal Process.*, Jakarta, Indonesia, 2014, pp. 1–6.
- [151] C. Yang, G. Cheung, and V. Stankovic, "Estimating heart rate via depth video motion tracking," in *Proc. IEEE Int. Conf. Multimedia Expo.*, Turin, Italy, 2015, pp. 1–6.

- [152] H. Padole, S. D. Joshi, and T. K. Gandhi, "Graph wavelet-based multi-level graph coarsening and its application in graph-CNN for alzheimer's disease detection," *IEEE Access*, vol. 8, pp. 60 906–60917, Mar. 2020.
- [153] L. T. Xuyen, L. T. Thanh, N. L. Trung, T. T. T. Quynh, and N. D. Thuan, "Eeg source localization: A new multiway temporal-spatial-spectral analysis," in *Proc. 6th NAFOSTED Conf. Inf. Comput. Sci.*, Hanoi, Vietnam, 2019, pp. 228–233.
- [154] X. Ni, T. Gao, T. Wu, J. Fan, and C. Chen, "Learning human cognition via FMRI analysis using 3D CNN and graph neural network," in *Multimodal Brain Image Analysis and Mathematical Foundations of Computational Anatomy*. Cham, Switzerland: Springer, 2019, pp. 93–101.
- [155] N. Ghoroghchian, D. M. Gropp, R. Genov, T. A. Valiante, and S. C. Draper, "Node-centric graph learning from data for brain state identification," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 6, pp. 120–132, Jan. 2020.
- [156] Y. Zhang and H. Huang, "New graph-blind convolutional network for brain connectome data analysis," in *Proc. Int. Conf. Inf. Process. Med. Imag.*, 2019, pp. 669–681.
- [157] Q. Lian, Y. Qi, G. Pan, and Y. Wang, "Learning graph in graph convolutional neural networks for robust seizure prediction," *J. Neural Eng.*, vol. 17, no. 3, May 2020, Art. no. 035004.
- [158] Y. Seo, M. Defferrard, P. Vandergheynst, and X. Bresson, "Structured sequence modeling with graph convolutional recurrent networks," in *Proc. Int. Conf. Neural Inf. Process.*, 2018, pp. 362–373.
- [159] B. Baingana and G. B. Giannakis, "Tracking switched dynamic network topologies from information cascades," *IEEE Trans. Signal Process.*, vol. 65, no. 4, pp. 985–997, Feb. 2016.
- [160] V. Kalofolias, A. Loukas, D. Thanou, and P. Frossard, "Learning time varying graphs," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, New Orleans, LA, USA, 2017, pp. 2826–2830.
- [161] P. J. Mucha, T. Richardson, K. Macon, M. A. Porter, and J.-P. Onnela, "Community structure in time-dependent, multiscale, and multiplex networks," *Science*, vol. 328, no. 5980, pp. 876–878, 2010. [Online]. Available: <http://www.jstor.org/stable/40655926>
- [162] J. S. Stanley, E. C. Chi, and G. Mishne, "Multiway graph signal processing on tensors: Integrative analysis of irregular geometries," *IEEE Signal Process. Mag.*, vol. 37, no. 6, pp. 160–173, Nov. 2020.



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